Chapter 2

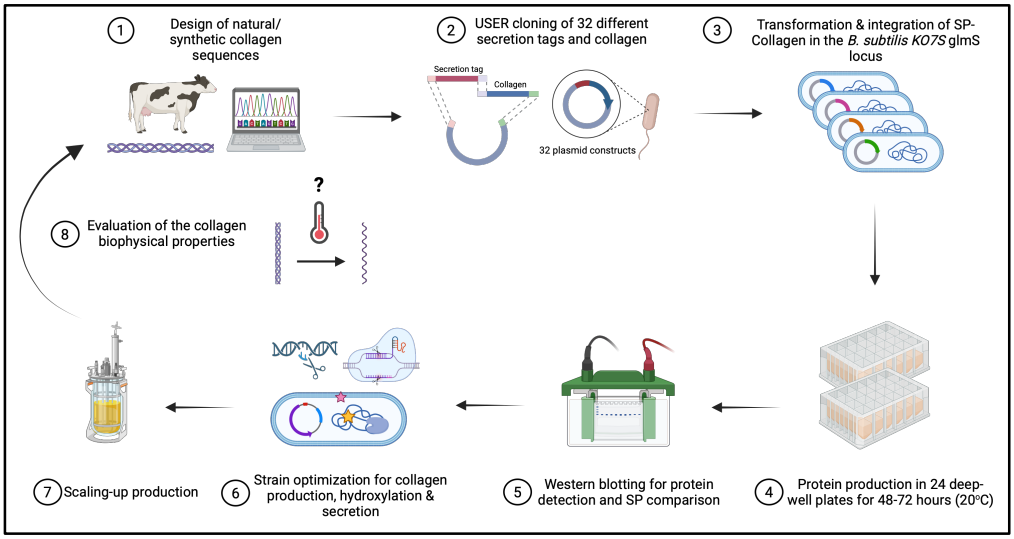
Production & secretion of hydroxylated bovine collagen and other polymeric proteins in *Bacillus subtilis*

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Abstract

Collagen is a polymeric protein with unique functional properties, including high water-binding capacity, gel-forming ability, and emulsifying potential, making it widely used in the food and pharmaceutical industries. However, traditional animal-derived extraction methods raise ethical, environmental, and safety concerns. Precision fermentation offers a sustainable alternative, though major challenges remain in efficient secretion and proline hydroxylation, which are essential for collagen's structural and functional stability. In this study, we report the successful recombinant production and secretion of bovine collagen, and other polymeric proteins in *Bacillus subtilis*. By constructing and screening a library of 32 secretion tags, we identified several signal peptides that enabled efficient extracellular secretion of codon-optimized collagen fragments. Additionally, by extending the size of collagen constructs - from short bioactive sequences to full-length tropocollagen – we revealed the size-dependent limitations of collagen production, particularly in sequences containing the native C-terminal propeptide. To introduce proline hydroxylation, we co-expressed prolyl-4 hydroxylases (P4Hs) from *Bacillus anthracis* (BaP4H) and *Giant Mimivirus* (MimiP4H). Proteomic analysis indicated limited hydroxylation by BaP4H, while MimiP4H exhibited promising results, with >30% of collagen-derived peptides showing hydroxylation in intracellular fractions. We confirmed by using circular dichroism, the formation of triple-helical structures for all constructs, with a similar melting temperature of 29.9 °C, regardless of hydroxylation status. Furthermore, we extended the secretion platform to other structural proteins, including resilin and elastin. To our knowledge, this study represents the first successful production of hydroxylated collagen and resilin in *B. subtilis*, laying the groundwork for sustainable production of protein-based biopolymers for applications in food, materials, and biomedical industries.

Introduction

Collagens are the most abundant proteins in the animal kingdom, accounting for approximately one-third of the total protein content. The collagen superfamily consists of 28 members in vertebrates (I – XXVIII) and all share a common structural motif, the triple-helix[1–3]. This supramolecular structure is composed of three polypeptide chains, stabilized by hydrogen bonds and the presence of the uniquely modified amino acid

hydroxyproline (Hyp – O)[4,5]. Each chain is composed of repeating Gly-X-Y sequences, where X and Y are often proline and hydroxyproline, respectively[6,7]. This arrangement provides collagen with characteristic tensile strength and thermal stability[6]. As a structural member of the extracellular matrix, it plays a crucial role in providing mechanical strength and resilience to connective tissues, while also promoting cell adhesion, differentiation, and growth[8–11].

Beyond its physiological role, collagen and its derivative, gelatin, exhibit unique properties, such as high water-holding capacity, gel-forming ability and emulsifying potential[12]. These attributes make collagen a coveted protein for applications in the pharmaceutical, cosmetic and food industries[12–19]. As of 2025, the global market for collagen and gelatin was valued at over 10 billion USD, with strong growth driven by the demand for novel functional foods and peptide health supplements[20]. However, the escalating demand for collagen-based products could be limited by the insufficient amount of current primary material sources, failing to meet the projected needs[21].

Traditionally, collagen has been extracted from animal tissues and bones, primarily from bovine and porcine origins. However, during the last few decades, there has been a significant shift away from these resources[12,22]. Animal agriculture has long been a key primary protein source for the global population. Nonetheless, it is now recognized as a significant contributor to greenhouse gas emissions and land use, both of which are central drivers of the current climate crisis[23–25]. Additionally, the consumption of bovine or porcine-derived products is restricted due to religious or ethical constraints, and isolating native collagen involves expensive extraction methods, batch-to-batch variations and the risk of transmissible diseases[26–28]. Hence, there is an urgent need for novel alternative sources of protein, including collagen[29,30].

Microbial fermentation has emerged as a compelling strategy for producing recombinant collagen and collagen-like proteins[31,32]. Unlike traditional methods, precision fermentation allows the controlled production of safe, high purity, and reproducible biomolecules using genetically engineered microorganisms[33]. Over the past decades, recombinant collagen has been produced with varying degrees of success in a diverse set of host organisms that include bacteria, yeast, insect cells, plant cells and transgenic plants[33,34]. Most of the efforts have suffered from low titers while the usual intracellular

production of recombinant collagen significantly increases the operating costs, hindering the economic viability of commercial-scale production[32,34]. Additionally, a critical barrier remains the inefficient post-translational modification of proline to hydroxyproline, a reaction that is catalyzed by prolyl-4 hydroxylase (P4H) which is essential for achieving the stable, functional structure of native collagen stable[35]. Expression of recombinant animal P4Hs demonstrated low levels of solubility or enzymatic activity. However, evidence suggests the presence of P4Hs in bacteria, algae, and viruses, could exhibit similar functions as animal P4Hs[36,37]. Specifically, *Bacillus anthracis* P4H (BaP4H) and *Giant Virus Mimivirus* P4H show higher hydroxylation efficiency when compared to their animal counterparts when expressed in *E. coli*[38–41]. However, it is uncertain whether similar activity can be achieved in other bacterial hosts.

Among bacterial platforms, *Bacillus subtilis* stands out as a promising chassis for recombinant protein production due to its GRAS (generally regarded as safe) status, high protein secretion capacity, and robust genetic tools[42,43]. Unlike hosts with limited secretion capabilities, like *E. coli*, *B. subtilis* naturally secretes proteins into the extracellular medium, simplifying downstream processing and enhancing cost-effectiveness. These traits make it an ideal host to produce complex proteins, like collagen. However, the efficiency of protein secretion is dependent on the interplay between the secretion tag and the protein of interest. So far, no accurate prediction tools have been developed, making the experimental exploration of the appropriate tag essential[44].

In this study, we present the first successful production and secretion of hydroxylated recombinant collagen in *Bacillus subtilis.* To identify the best secretion signals for collagen, we integrated a library of secretion peptides into the genome of an engineered *B. subtilis* strain and conducted a systematic secretion assay. Based on these results, we developed integrative and expression vectors enabling the production and secretion of higher molecular-weight polymeric and collagen proteins. Additionally, we explored the hydroxylation effects of two P4Hs on the recombinant collagen and assessed its impact on the biophysical properties of collagen, such as triple-helix formation, which proved to be stable at 29.9°C regardless of the hydroxylation status. Our findings establish the foundation for an animal-free collagen production, paving the way for future applications in food technology.

Results and discussion

1. Efficient production and secretion of non-hydroxylated bovine collagen

Although *Bacillus subtilis* has been widely used in industry for the secretion of heterologous proteins and enzymes, the efficiency of secretion is dependent on the interplay between the secretion tag (SPs) and protein sequence of interest (POI).

To establish an efficient secretion chassis for recombinant bovine collagen, a 110 amino acid sequence of bovine collagen type I alpha I (10kDa) was selected, codon-optimized and cloned with a library of 32 N-terminal secretion tags. The secretion tag library was constructed by amplifying native *Bacillus subtilis* sequences, while the collagen fragment was carefully chosen to (a) reflect the amino acid composition of full-length collagen (**Supplementary Table 1**), (b) originate from a region predicted to form a stable triple-helix, and (c) include putative bioactive motifs such as VPGDL and the integrin α2β1-binding sequence DGEA[45–48] . The 10kDa collagen was named BsbColv2.

To achieve comparable results, integrative plasmids targeting the *glmS* locus of the *Bacillus subtilis* genome were employed. Expression of collagen constructs was driven by the strong constitutive promoter P3P[49].

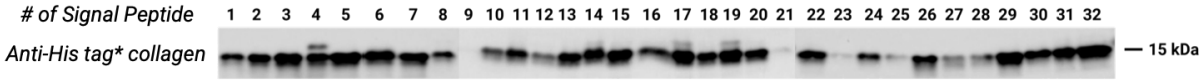


Figure 2 | Western blotting of secreted his-tagged collagen. *Each lane represents the culture supernatant of different collagen secreting strains*

Western blot results indicated that most strains were able to effectively produce and secrete collagen in the production media (**Figure 2**). Although the expected molecular weight (MW) of the collagen fragment is 10kDa, the bands showed reduced SDS-PAGE mobility, which corresponds to a higher MW (15kDa). This phenomenon is common for collagen and other collagenous proteins[50]. The most prominent strains were selected by comparing the intensity of the bands, normalized by the estimated total protein content. The experiments were performed in triplicates. Each secretion tag was compared directly to other tags in the same gel (**Figures 3A-C**).

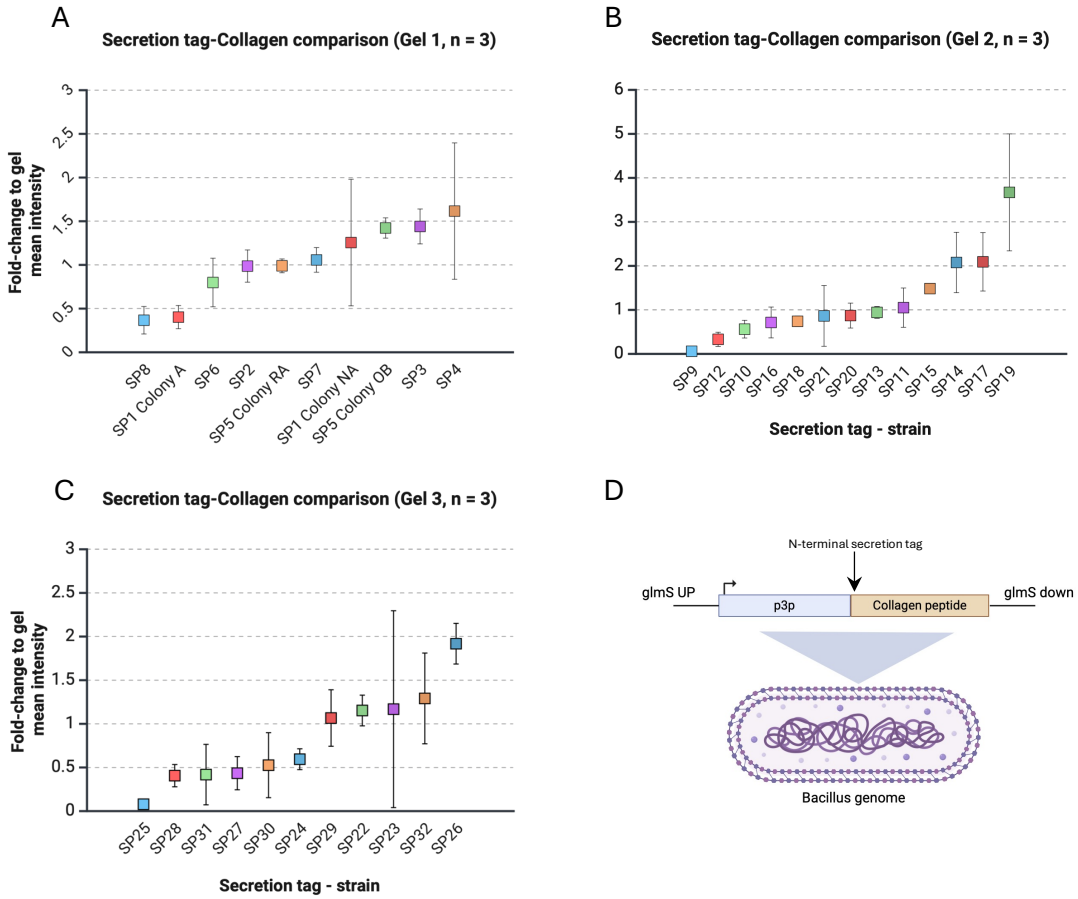


Figure 3* | *Quantification of secreted collagen levels per secretion tag. *A-C, The intensity of the western blot bands was measured and normalized by estimated total protein per lane. The mean intensity per gel was used to estimate the fold-change of each construct. Each graph represents a different gel. The experiments were repeated at least three times, from production to westen blotting. D. Graphical representation of the genomic expression of collagen with N-terminal secretion tag.*

Unexpectedly, some of the strains produced consistently a secondary band at a higher molecular weight (**Figure 2**). This could be due to insufficient cleavage of the secretion tag or resistance to a potential degradation event. Notably, our selected collagen amino acid sequence lacks sufficient aromatic and charged amino acids to enable Coomassie-based visualization. Consequently, in-gel mass spectrometry analysis was not feasible, and the identity of the secondary band remained elusive.

To validate the best performers, an additional experiment with six of the best candidate secretion bacterial strains was completed. These correspond namely to SP3, SP4, SP5, SP19, SP26, S32 strains. The strain with SP2 secretion tag was included as an internal control of mid-level secretion (Fold change equal to 1).

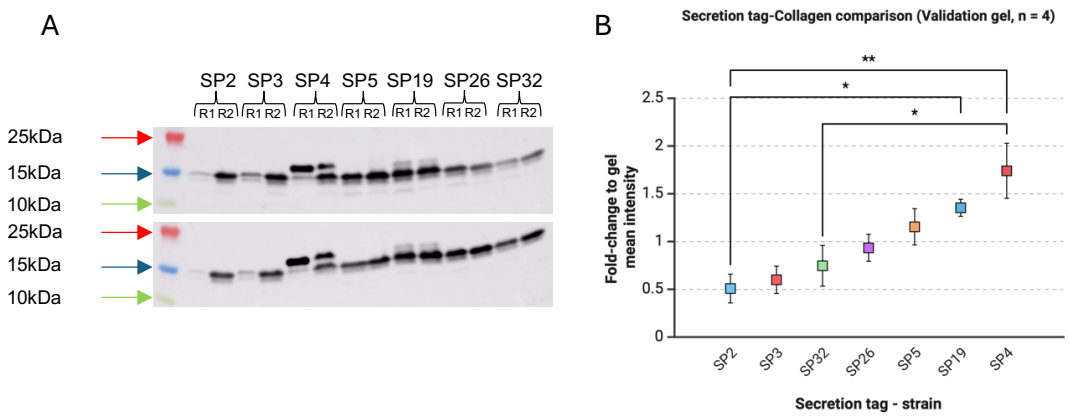


Figure 4 | Quantification of secreted collagen levels per secretion tag. *The intensity of the western blot bands was calculated as before. The strains SP4 and SP19 showed significanlty increased secretion efficiency compared to SP2 internal control. SP4 (Figure 4A) consistently produced a secondary band, with lower mobility in the SDA-PAGE. The results were in biological and technical replicates.*

As shown in **Figures 4A** and **4B,** every strain has outperformed SP2 strain, validating our previous results and the superiority of our selected secretion tags. Specifically, the SP4 strain showed statistically significantly elevated levels of secretion compared to strains SP32, SP3 and SP2. Interestingly, SP4 is the only strain that provides consistently a secondary band of higher MW. To avoid the production of multiple proteins simultaneously and maintain the reproducibility of the recombinant collagen, we decided not to include SP4 secretion tag in later experiments, despite its superior secretion. In conclusion, the signal peptides SP19, SP5, SP26 and SP3 were selected for further experiments.

2. Expanded pool of secreted collagen α1[I] sequences and polymeric proteins

To explore if the selected secretion tags can translate similar results to polymeric proteins of greater molecular weight, we decided to evaluate larger sequences stemming from bovine collagen type I, including full-length collagen and procollagen. Additionally,

we expanded the application of the strain to production of other protein-based biopolymers, such as elastin and resilin.

Progressive extension of the codon-optimized collagen sequence size

Based on literature review, we found out that a region close to our initial 10kDa recombinant collagen contained additional bioactive domains, including domains that control fibroblast adhesion and exhibit antioxidant activity. As these properties could prove beneficial in food applications, we included them in a newly designed 20kDa recombinant collagen (named Bacola225). The western blot results (**Figure 5B**) showed that the 20kDa collagen was successfully produced and secreted by all the secretion tags, with minor fluctuations in secretion efficiency. When compared to the previously produced 10kDa fragment (PC), the novel collagen shows similar or minor decreased titers.

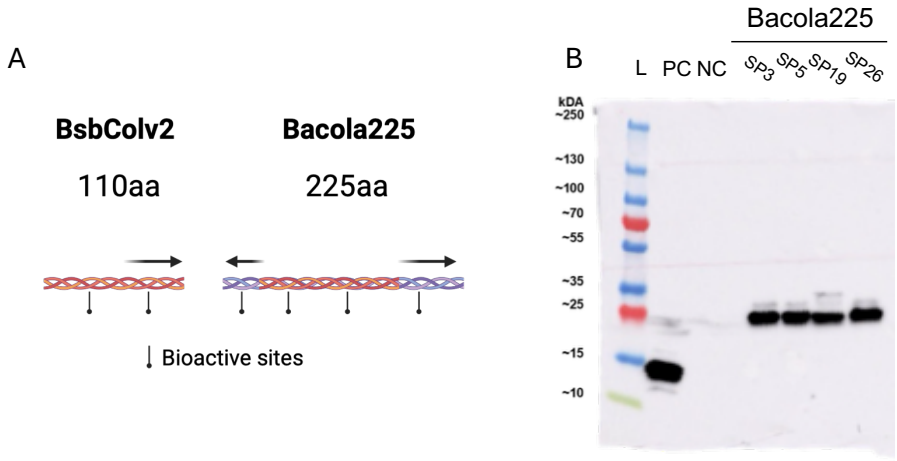


Figure 5 | Contruction and production of Bacola225*. A. Graphical representation of the extension from 110aa (BsbColv2) to 225aa (Bacola225). Bacola225 contains additional bioactive sites. B. Western blotting of hig-tagged Bacola225 (C-terminal). L: Ladder, PC: Positive control – BsbColv2 (SP5), NC: Negative control – Supernatant of KO7S.*

To evaluate the effect of increasing molecular weight on collagen's production, we designed extended constructs by doubling the codon-optimized sequence of collagen twice, while maintaining the amino acid sequence of native collagen. Therefore, our pool of collagens includes proteins of 110, 225, 450 and 900 amino acids that correspond to 10, 20, 45 and 87kDa **(Figure 6).** To build the constructs, we extended the previously produced collagens towards the C-terminal by Gibson assembly and integrated them to the *glmS* site of *Bacillus subtilis* as previously. For this study, we restricted our constructs to include only secretion tag SP5. Unfortunately, the results showed that the production

for the 450 amino acid construct was limited only in the intracellular fraction and in insufficient amounts while the production and secretion of the 900 amino acid construct was not possible (**Figure 6 and Supplementary Figure 1**). It should be noted, that the 900aa construct included the C-propeptide of collagen α1[I]. It is believed that C-propeptide promotes collagen solubility in native conditions, however it seems - from our data - that it might hinder the production of recombinant collagen in *Bacillus subtilis* systems.

The idea behind this substudy of progressive extension was to uncover putative mechanisms of stress, that are specific to collagen, with the employment of transcriptomic analyses. However, the limited to non-existent production of larger collagens was a significant factor in not pursuing further analysis. However, we conclude that the extensive resources required by expanded collagens, considerably hinder and stress *B. subtilis* production capabilities, is a size dependent manner.

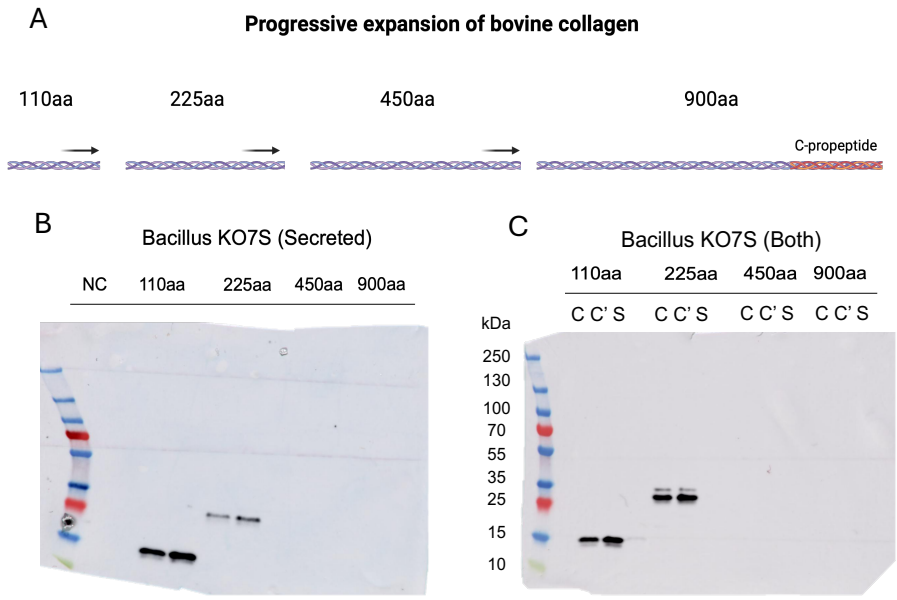


Figure 6 | Contruction and producion of progressively extented collagens. *A. Graphical representation of doubling extension steps. Bacola900 contains the a1[I] C-propeptide. B, C: Western blots of his-tagged constructs. B, corresponds to media supernatant and C to both replicates of cell pellets (C, C') and S supernatant.*

The C-propeptide affects the secretion of α1[I] and α2[I] collagen chains

Furthermore, we focused on other regions that control the assembly of collagen. In nature, collagen type I is composed of a 2:1 ratio of alpha 1 (a1[I]) and alpha 2 (a2[I])

chains. The C-terminal propeptide, also known as the C-propeptide, plays a crucial role in collagen assembly and fibril formation both inside and outside cells. It guides the selection and association of collagen alpha chains, ensuring proper folding and preventing premature fibril formation within cells. Once the procollagen molecule is secreted, the C-propeptide is cleaved by specific native proteinases, leading to the release of the mature collagen molecule and subsequent fibril assembly. To investigate the manifestation of similar effects, we cloned the codon-optimized C-terminal sequences of both alpha I and alpha II chains with and without their respective C-propeptides. We found out that the secretion of alpha I chain with the C-propeptide was possible for three out of the four secretion tags, with peptide SP26 being the most efficient (**Figure 7C**). On the contrary, the alpha 2 chain sequence that maintains the C-propeptide, was not able to be secreted (**Figure 7C**)**.** Further experiments showed that limited amounts of alpha 2 collagen are produced and retained inside the bacterial cells (**Supplementary Figure 2**). Notably, the secretion of alpha 1 and alpha 2 sequences lacking their respective C-propeptides yields different results. Both chains are able to be secreted in the production media in relatively similar amounts (**Figure 7D**). Particularly, the consistency of secretion irrespectively of the presence of the C-propeptide or not, renders secretion tag SP26 an ideal partner for production of the α1[I] chain. Although the two chains are designed to have similar molecular weight, the electrophoretic mobility of α2 chain is higher than that of α1, a phenomenon that aligns with that of native α1[I] and α2[I] chains[51]. Interestingly, a single degradation band is visible for both chains, indicating a precise degradation pattern, possibly by an unknown host-specific protease. When compared against the previous 10kDa collagen, the production was visibly significantly lower.

Successful production of full-length tropocollagen from bovine cDNA

As a next step, we explored the production of procollagen (138kDa) and tropocollagen type I alpha I (96kD) – full length collagens with and without their N- and C- terminal propeptides, respectively. The polymeric and high in GC% collagen sequence does not allow the production of long sequences from DNA synthesis companies, even with extensive codon-optimization. For this reason, we decided to amplify the bovine procollagen and collagen α1[I] sequences directly from purchased bovine skin cDNA. As with previous constructs, we integrated the genes and their respective secretion tags to

the *glmS* locus of *Bacillus subtilis* KO7S strain and tested their production. Interestingly, it was revealed that the production and secretion of tropocollagen is possible in *B. subtilis* for two out of four secretion peptides – SP5 and SP19, albeit in smaller amounts (**Figure 7F**) However, this is an important milestone, as the number of successfully secreted proteins >90 kDa is limited in academic literature. By increasing the total volume loaded in SDS-PAGE, we found out that the production of tropocollagen is possible from the other secretion tags as well, but with a significant loss in titers (**Figure 7G**). Furthermore, it was apparent that thaw-freeze cycles or proteolytic activity produced degradation products (**Figure 7G**). On the contrary, procollagen was not produced in either intracellular or extracellular fraction, demonstrating that potentially sequences in N- or C-terminal propeptides hinder collagen production. Nevertheless, the expression and production of a full-length collagen from a mammalian cDNA remains remarkable as the sequence is not codon-optimized and *Bacillus subtilis 168* does not exhibit high codon usage for codons like CCC which is prominent in collagen sequence and translates to proline.

To investigate if codon-optimized versions could prove advantageous for the production of procollagens α1[I] and α2[I], we assembled synthetic constructs by USER cloning codon-optimized segments to approximate the full-length sequences, while maintaining the native amino acid content. These constructs were integrated into the *glmS* locus and paired with the validated SP5 secretion tag as before. Western blotting showed that procollagen was produced in small amounts, but it was maintained inside the cell (**Supplementary Figure 3**). Future experiments will aim on testing more signal peptides to achieve secretion and compare the production of codon-optimized tropocollagen to the cDNA-derived tropocollagen.

No.	Name by Scietist	Expected secreted kDa*	AA Length	PIs	Source	Codon-Optimised
1	BsbColv2	9.9 kDa	108	6.3	Bovine a1[I] collagen	Yes
2	Bacola225	20.3 kDa	225	8.3	Bovine a1[I] collagen (contains BsbColv2)	Yes
3	Bovine cDNA Collagen I	95.5 kDa	1063	9.3	Bovine a1[I] collagen without N, C propeptides	Native
4	a1[I] with C propeptide	48 kDa	470	6.21	Bovine a1[I] collagen	Native
5	a1[I] without C propeptide	20.4 kDa	221	7.29	Bovine a1[I] collagen	Yes
6	a2[I] without C propeptide	20.6 kDa	222	9.69	Bovine a2[I] collagen	Yes
7	Resilin	56.8 kDa	604	5.15	*D. melanogaster*	Yes
8	Elastin	68. 4 kDa	786	10.40	Human	Yes

Table 1 \| Overview of successfully produced and secreted proteins in this part of the *study. Information about the expected MW of protein, amino acid length, source, pH isoelectric point and subjection to codon-optimization.*

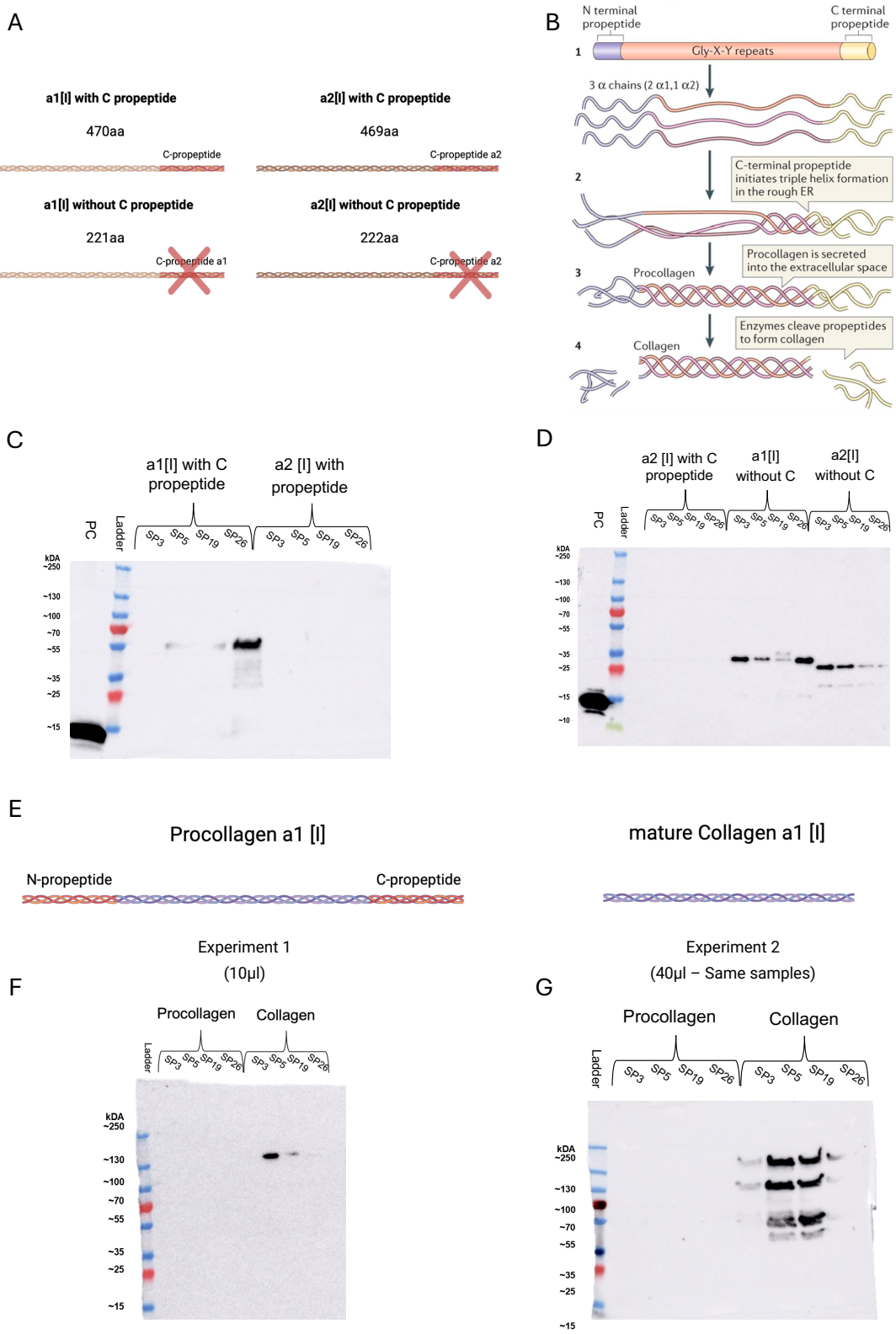


Figure 7 | Graphical representations and westen blots of α1[I], α2[I] collagen fragments, full tropocollagen and procollagen. *A. Overview of collagen fragments with and without their respective C-propeptide. B. C-propeptide is important for the nucleation and triple-helix formation. C, D: Western blots of collagen fragments with and w/o their respective C-propeptide. Production is affected by C-propeptide. E. Oveview of Procollagen and full collagen. F, D Western blots of the same samples against his-tagged Procollagen and collagen. 10µl or 40µl sample volume was used. PC: Positive control – BsbColv2 SP5*

Achieving production and secretion of other protein-based biopolymers: Elastin & Resilin

To assess the ability of *Bacillus subtilis 168* to produce other biopolymers, we decided to explore the production of other codon-optimized elastomeric proteins such as elastin and resilin. Similarly to collagen, these proteins are characterized by repeated, glycine- and proline-rich domains that confer elastic and self-assembly properties. Resilin, for example, is purported to be one of the most resilient elastic materials known[52]. Additionally, elastin often co-localizes with collagen in the extracellular matrix and 20-24% of its prolines are hydroxylated[53]. These shared structural and biochemical features make elastin a promising candidate for transferring insights we are gaining from the development of our collagen production platform and open the possibility of co-utilizing our biopolymers in future applications.

For this study, we employed the pHT100 plasmid that contains the IPTG-inducible pgrac100 promoter. pHT100 is a mid-copy plasmid and it has been described as a strong driver of protein production[54]. In a similar manner as described before, we assembled the biopolymers' sequence by USER cloning codon-optimized segments together (Figure 8A). SP5 was used again as secretion tag. However, the His tag was placed in the N-terminal directly after the secretion tag followed by a TEV cleavage site. After protein production experiments in Cal18-2, we tested the detection of elastin or resilin in either intracellular or secreted fractions. Western blots revealed the efficient secretion of both resilin and elastin in the production media, with limited amounts maintained in the cell lysates (**Figure 8**). Notably, the secretion of elastin, proved to be inconsistent between batches, with some experiments revealing the absence of elastin secretion (**Supplementary Figure 3**). On the contrary, resilin secretion proved to be more robust leading to successful purification in a larger batch. Interestingly, during dialysis on TBS at 4°C, the resilin solution turned opaque (**Supplementary Figure 4B**). This phenomenon is characteristic of native resilin as the protein's conformation is pH- and temperature-dependent and exhibits an upper critical solution temperature (UCST) around 6°C (**Supplementary Figure 4A**)[55]. Temperatures below the UCST, promote conformational changes that lead to phase separation and visible opaqueness. This suggests that our recombinant variant might mimic the native resilin's structure. Further experiments are required to characterize the recombinant resilin and explore its potential applications.

Nevertheless, to the best of our knowledge, this is the first time that resilin has been produced and secreted by a *Bacillus subtilis* strain.

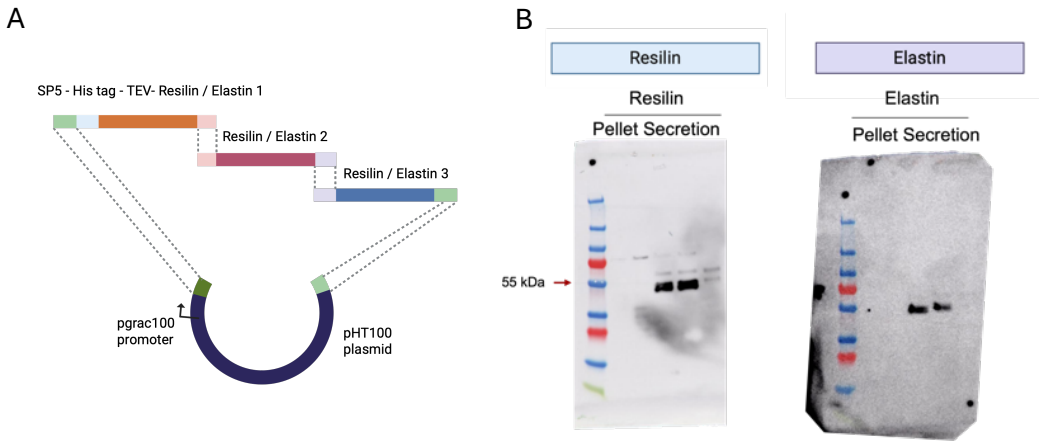


Figure 8 | Cloning and western blotting of recombinant resilin and elastin. *A. DNA fragments obtained from DNA synthesis were amplified with Uracil-containing primers and assemblied in the pHT100 plasmid with USER cloning. B. Western blotting of recombinant resilin and elastin reveals successful secretion of both.*

3. Hydroxylation of recombinant collagen by non-animal *prolyl-4 hydroxylases*

Bacillus anthracis P4H exhibits low levels of expression and hydroxylation activity

The post-translational modification of proline to hydroxyproline by prolyl-4 hydroxylase is crucial for the stability of collagen. To achieve collagen hydroxylation in *Bacillus subtilis*, we employed *B. anthracis* P4H (BaP4H), known to act on collagen-like sequences, and evaluated the enzymes activity on recombinant collagens *in vivo*. We constructed two different systems for the co-expression of collagen and BaP4H. The first one builds on the previously developed strains that constitutively express the 10kDa and 20kDa collagen genes coupled with SP5 signal peptide. These strains were transformed by natural competence with a ProUSER plasmid, that bears homology arms for the *rsbP* locus and the BaP4H codon-optimized gene, controlled by the Tet-inducible promoter. To monitor BaP4H production, we added a N-terminal Strep-tag II. The enzyme has a dimerization domain close to the N-terminal and an active site in the C-terminal. Acknowledging that the Strep-tag might be affected by this conformation, we decided to develop an additional construct and including a GS linker (spacer) between the Strep-tag and the rest of the P4H sequence (**Figure 9A**). The second system represents an operon system of BaP4H and the respective collagen (**Figure 9B**). For this, we used the pHT01

plasmid that includes the IPTG-inducible pgrac01 promoter. Western blot analysis verified the production of collagen from both systems. It indicated that the production of collagen from the operon system is more consistent when induced by 0.5mM IPTG (**Supplementary Figure 5**). Additionally, the detection of Strep-tagged BaP4H was not possible, however, Ponceau S staining provided indications of BaP4H presence inside the cells (presence of distinct bands – **Supplementary Figure 6**).

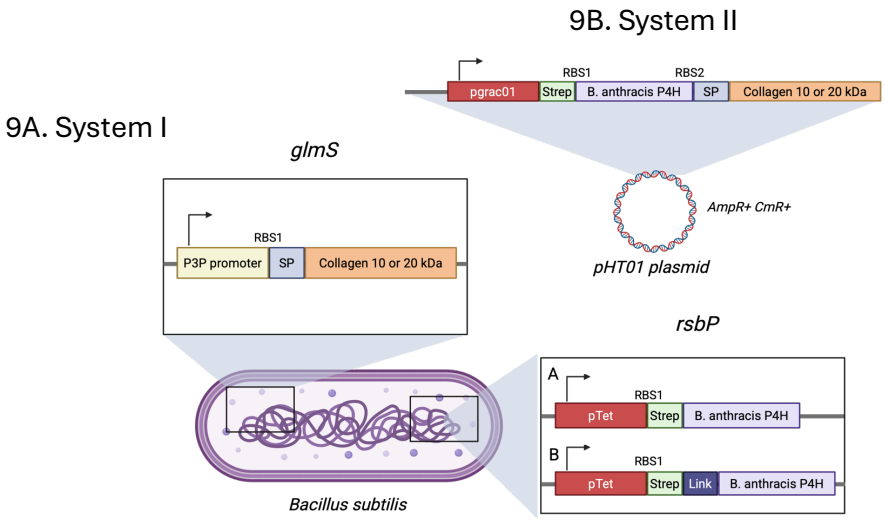


Figure 9 | Co-expression systems of BaP4H and collagen fragments. *9A. This system is based on chromosomic expression of collagen as before, and integration of Tet-incublible BaP4H in the rsbP locus. BaP4H has a Strep tag II in N-terminal, with or without a GS link inbetween. 9B system uses the IPTG-inducible pHT100, to express BaP4H and collagen fragments from an operon.*

To investigate if BaP4H was produced and active, we performed proteomic analysis on the cell pellets and on selected purified collagens (secreted fractions). For comparison, we included *E. coli* strains that express both BaP4H and the recombinant collagens (not included in this thesis). Unexpectedly, the BaP4H was detected only in samples of the operon system. Compared to *E. coli*, which achieved BaP4H expression levels exceeding 6.2×10^{11} intensity units, *B. subtilis* exhibited significantly lower expression, reaching only around 3×10^{6} units—representing a difference of over five orders of magnitude (**Figure 10A**). Of the purified secreted collagens, detectable hydroxylated peptides were present in only 3 samples and include only 1 hydroxylated site per peptide (**Figure 10b**). These samples include: the Operon-BaP4H-Collagen110, the *glmS*::Collagen110 *rsbP*::NoLinkerBaP4H and the *glmS*::Collagen225 *rsbP*::LinkerBaP4H. Although the coverage of collagens was 100% and 73% for the 10kDa and 20KDa collagens

respectively, the absence or limited detection of hydroxylated peptides indicates that hydroxylated fraction may be obscured by the non-hydroxylated counterparts indicating that the occupancy rate of hydroxylated prolines is generally low. Thus, we conclude that neither system as it stands, can facilitate efficient production of hydroxylated collagen. We hypothesized that the low hydroxylation levels are a function of the low abundance of BaP4H and the limited availability of cofactors. Most prolyl-4 hydroxylases need oxygen, a-ketoglutarate, iron and ascorbate to fully function. For this reason, we decided to increase the mRNA levels of BaP4H, supplement the media with ascorbate and explore the use of an additional non-animal P4H.

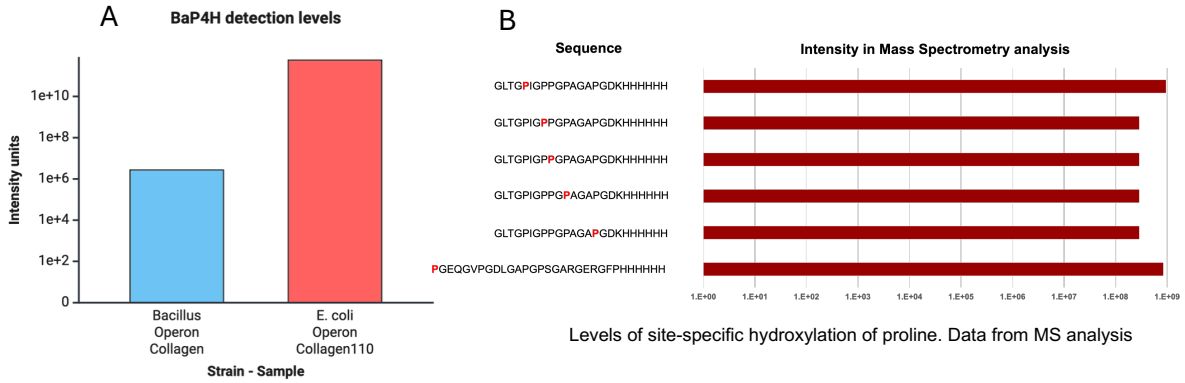


Figure 10 | First Proteomics results for detection of BaP4H and hydroxylation on purifed collagens*. Comparison of detected BaP4H between Bacillus subtilis pHT01 operon and E.coli T7 system. The levels of BaP4H in Bacillus were detected only in the Operon P4H – BsbColv2 and are magnitutes lower that what E.coli can reach. B. Hydroxyproline was detected in recombinant collagen, albeit peptides bearing only 1 event of hydroxylation, indicate low activity of P4H.*

Optimization of P4Hs expression and hydroxylation conditions

To optimize P4H and collagen expression, we focused on the operon system, producing only Bacola225 (20 kDa collagen). Initially, we employed the pHT100 plasmid to increase the mRNA levels through the pgrac100 IPTG-inducible promoter. Moreover, we increased the number of P4H tested, by including *Mimivirus* P4H (MimiP4H) in our analysis. This viral P4H has been proven to effectively work on collagen expressed in prokaryotes, such as *E. coli.*

To monitor the enzymes production, we substituted the Strep-tag II in the N-terminal, with the Nano-Glo® HiBiT-tag, a constituent of a sensitive luciferase system for protein detection. Expression of P4Hs - without collagen - in *B. subtilis KO7S* strains, and subsequent Hibit-blotting experiments, revealed successful production and detection of

both enzymes (**Figure 11A**). However, the levels of Mimivirus P4H were significantly lower than that of the BaP4H, suggesting the need for increased translation.

The ribosomal binding site (RBS) is a significant factor controlling the production levels of proteins. Hence, we decided to explore and optimize the strength of our RBSs through the employment of RBS and Operon Calculator bioinformatic tools[56]. New designs increased considerably the estimated P4H translation rate from 289au to 241386au and the collagen translation rate from 102358 to 197692 (**Supplementary Figure 7**). To provide experimental validation, we aimed to construct plasmids bearing either new or previous RBSs for both P4H and the 20kDa recombinant collagen, with or without secretion tags. A significant increase in P4H or collagen titres would validate the predicted results. The new RBSs were namely RBS3* for P4H and RBS4* for collagen, instead of RBS1 and RBS2 as previously. The asterisks nominate their synthetic nature.

Constructs expressing the RBS1 and RBS2 for BaP4H and MimiP4H were successfully assembled. However, multiple cloning rounds failed to produce BaP4H-NewRBS constructs without disruptive mutations on the BaP4H gene. It is known that pgrac100 can be leaky in *E. coli*[54], and elevated BaP4H levels with collagen could be disruptive. Nevertheless, we managed to create operons for MimiP4H, that enable either intracellular or extracellular production of the 20kD collagen, and we set to explore the effect of RBS3* and RBS4* versus RBS1 and RBS2.

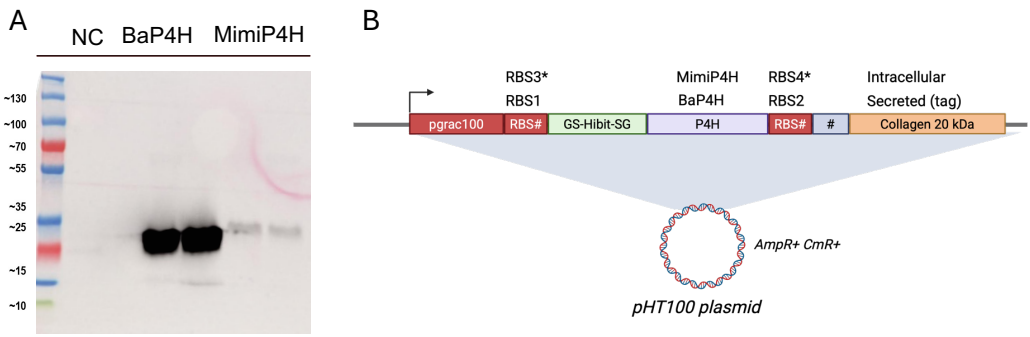


Figure 11 | Hibit-blotting for P4H detection and graphical representation of the new pHT100 plasmid combinations, with two P4Hs and synthetic RBSs. *A. Hibit-blotting was able to verify the production of BaP4H and MimiP4H. The latter had significant less expression. For this reason, new RBS were computationally designed and aimed to be constructed in new plasmids that are visible in B.*

As P4Hs require ascorbate as a cofactor, we also tested if external supplementation of ascorbate (Vitamin C) can improve production. Although crucial for P4H activity, ascorbate has been documented to inhibit *Bacillus subtilis* growth and

biofilm formation[57]. For this reason, we performed a Vitamin C dose-dependent growth assay. We found out that concentrations below 5mM do not affect the growth in exponential phase but decrease the final OD in a dose-dependent manner (**Supplementary Figure 8**). Acknowledging this, a concentration of 2mM Vitamin C was selected for subsequent experiments to balance between ascorbate availability and its inhibitory effect. We hypothesized that if supplementation of ascorbate is critical, we might observe visual effects on the electrophoretic mobility of – hydroxylated - collagen proteins.

As RBS3* and RBS4* were not cloned with BaP4H, we performed comparisons only between constructs of MimiP4H. Hibit blotting showed that RBS3* was significantly more efficient than RBS1, enhancing MimiP4H production (**Figure 12A**). Subsequent western blotting on the same membrane, did not produce visible changes in collagen production or secretion. Notably, although the production of intracellular collagen was almost non-existent, the intracellular fractions of the strains that secrete collagen maintained high levels intracellularly, possible due to a protective effect of the signal peptide (**Figure 12B**). A particular shift was observed between the secreted and intracellular fractions, that was attributed to signal peptide cleavage of the secreted collagen or collagen hydroxylation of the intracellular fraction. Simultaneously, no significant changes were observed due to treatment with ascorbate.

We also tested a genome-reduced *Bacillus* strain, PG10, which was reported to be beneficial for secreting difficult-to-express proteins[58]. Cross-comparison between the genome-reduced strain PG10 and KO7S, showed limited differences, except a minor increase in protein secretion with the RBS4* (SDA118 strain) and an additional band around 55kDa on the intracellular fraction of the RBS4* (SDA118). These are promising results for the utilization of the genome-reduced strains, however the strains showed inconsistent growth when cultivated in larger flasks, which is supported by well-documented differences in physiology compared to the wild strains[59,60].

As the next step, we verified with Hibit- and western-blotting, the production of BaP4H and secreted collagen respectively from strain SDA108 (RBS1 and RBS2 – **Supplementary Figure 9**) and constructed a novel strain (SDA130) that produces MimiP4H, and collagen intracellularly driven by the new RBSs (3* AND 4*). This strain was not included in the experiment in Figure 12. Interestingly, intracellular collagen production this strain (SDA130) provided a band in an unexpectedly higher molecular

weight, suggesting the possibility of hydroxylation and the formation of intracellular triple-helix. Additionally, small-scale IMAC purification of intracellular and extracellular collagen co-expressed with MimiP4H showed additional bands in a higher MW for the intracellular fractions, suggesting again the presence of hydroxylation and the potential association of collagens. This phenomenon was more prominent when samples were incubated at 4 ℃ overnight (**Supplementary figure 10**).

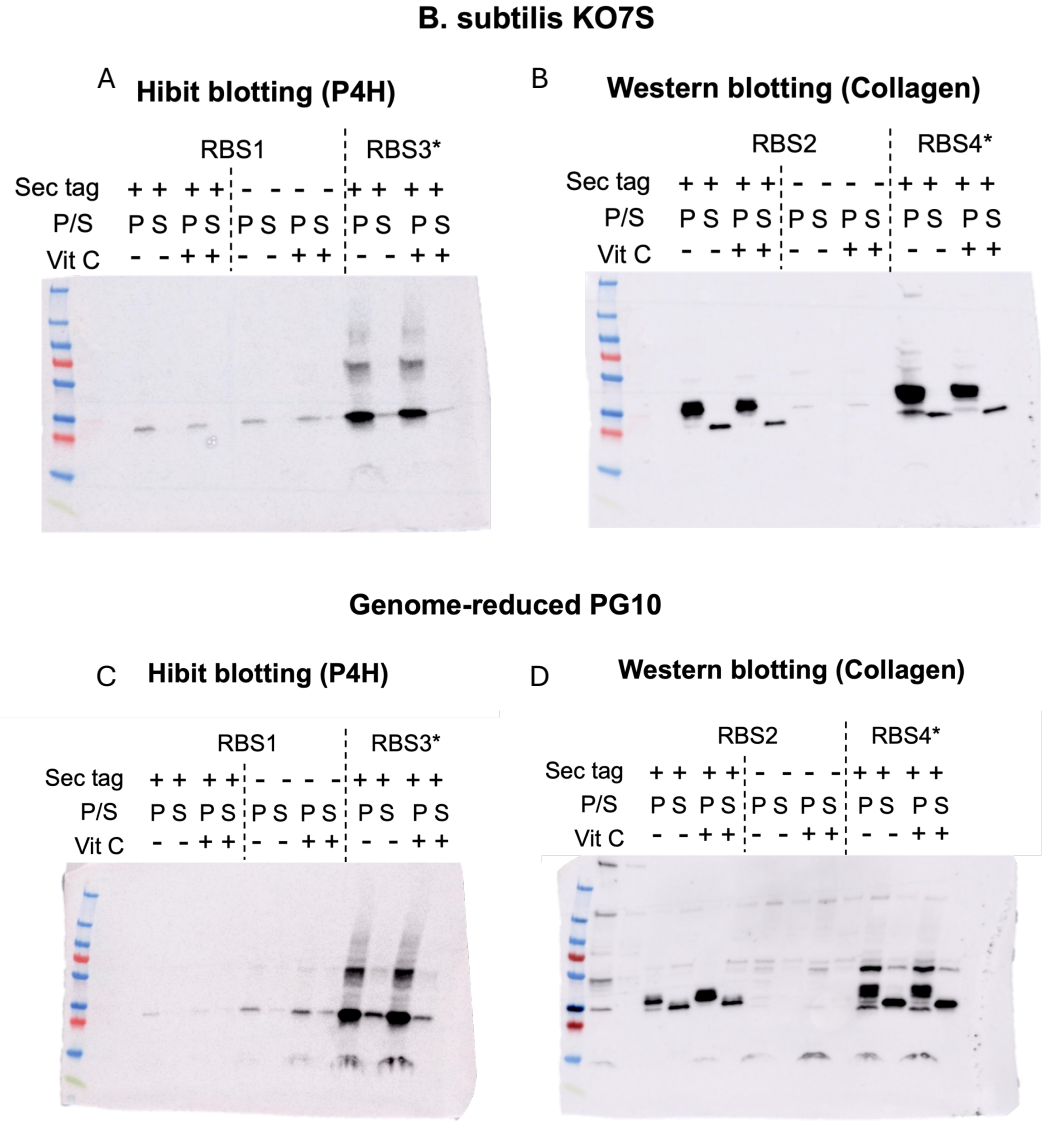


Figure 12 | Hibit and western blotting for comparison of operon producing MimiP4H and collagen with our in-house RBSs and the new synthetic RBSs, RBS3* and RBS4*. *Simultaneously, Vitamin C was supplemented to some instances without effect. RBS3* that controls the experession of the P4H has significantly increased P4H production. RBS4* showed minor increase in collagen production and secretion. Intracellular production was limited. Comparison between genome-reduced and KO7S, showed minor differences, such as increased collagen secretion in PG10. However, PG10 lacks growth in large flasks. This was performed in 24-wells.*

Name	Enzyme	RBSs	Production	Strain	Status
SDA108	*BaP4H*	*RBS1 & RSB2*	*Secretion*	*B. subtilis KO7S*	*OK*
SDA109	*BaP4H*	*RBS1 & RSB2*	*Intracellular*	*B. subtilis KO7S*	*No production*
SDA110	*MimiP4H*	*RBS1 & RSB2*	*Secretion*	*B. subtilis KO7S*	*OK*
SDA111	*MimiP4H*	*RBS1 & RSB2*	*Intracellular*	*B. subtilis KO7S*	*OK (Limited)*
SDA112	*MimiP4H*	*RBS3* & RBS4**	*Secretion*	*B. subtilis KO7S*	*OK*
SDA130	*MimiP4H*	*RBS3* & RBS4**	*Intracellular*	*B. subtilis KO7S*	*OK*
SDA116	*MimiP4H*	*RBS1 & RSB2*	*Secretion*	*Minibacillus PG10*	*OK*
SDA117	*MimiP4H*	*RBS1 & RSB2*	*Intracellular*	*Minibacillus PG10*	*OK (Limited)*
SDA118	*MimiP4H*	*RBS3* & RBS4**	*Secretion*	*Minibacillus PG10*	*OK*

***Table 2 | Strain information regarding the P4H, RBS, parent strain and production status**. RBS3* and RBS4* are novel synthetic RBSs, while RBS1 and RBS2 are the ones used before. Only one strain producing BaP4H was able to be constructed due to consistant instability and mutations during plasmid construction.*

4. Characterization of recombinant collagen with and without hydroxylation

4.1 Expression and purification of non-hydroxylated and hydroxylated collagen

Strain ID	P4H	RBS	Collagen	Collection
SDA091	-	RBS1	Bacola225	Intracellular & extracellular
SDA108	BaP4H	RBS1 & RBS2	Bacola225	Intracellular & extracellular
SDA112	MimiP4H	RBS3* & RBS4*	Bacola225	Intracellular & extracellular
SDA130	MimiP4H	RBS3* & RBS4*	Bacola225	Only intracellular

Table 3 | Table of strains used for characterization of collagen.

To assess the hydroxylation levels and structural stability of recombinant collagen, the production of non-hydroxylated collagen (SDA091), BaP4H-collagen (SDA108) and MimiP4H (secreted [SDA112] and intracellular [SDA130]) was further scaled up in 2L flasks containing 0.5L of terrific broth, supplemented with ascorbate and chloramphenicol.

To estimate the expected production of collagen from secretion or intracellular production in the case of SDA130, we performed a production batch and measured the OD_{600} and collagen production at specific time points. All strains manifested similar

growth patterns and reached approximately an OD_{600} of 10 (**Figure 13A**). The estimated collagen production, measured by ELISA, at time points of 2, 5 and 24 hours after induction showed significant differences between the - putatively - hydroxylated and non-hydroxylated constructs (**Figure 13B**). Secreted fraction of non-hydroxylated collagen (SDA91) reached 6mg/L while the ones that express *P4Hs* reached 1.8mg/L. Notably, the production of intracellular collagen (SDA130) was around 7.88µg/L or 0.829 ng/OD at 24 hours, verifying previous results of low intracellular titres. These results reveal significant barriers in efficient collagen production and demonstrate the need for further strain optimization and controlled fermentation in bioreactors to achieve meaningful titers. To collect the necessary collagen for future characterization experiments, multiple rounds of production and IMAC purifications were performed, isolating collagen both from culture supernatant as well as from the cell lysate (see Methods and **Supplementary Figure 11**).

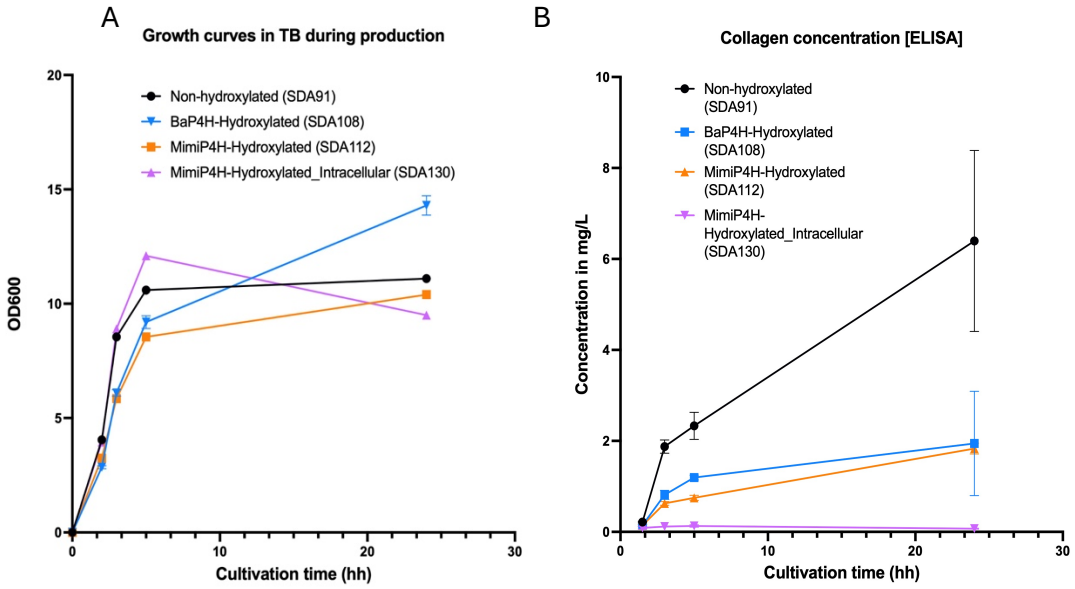


Figure 13 | Growth curves and collagen concentration measured with ELISA against his-tag. *Cultivation was performed in Terrific broth, with ascorbate and antibiotic. All strains grew fast in Terrific broth reaching an OD above 10, without significant differences. Secreted collagen titres were different for the strains that contain P4Hs compared to the control of only producing collagen. Intracellular production of collagen on strain SDA130, was significantly low.*

4.2 LC-MS/MS revealed significant MimiP4H activity and low hydroxylation in secreted collagen

A key challenge in microbial collagen production is achieving efficient proline hydroxylation. Provided that hydroxylation occurs intracellularly before secretion, we

hypothesized that collagen localized inside the cells could exhibit higher hydroxylation levels compared to secreted collagen molecules. This assumption is based on the idea that secretion, especially through the Sec pathway, may limit the exposure time of collagen to prolyl-4-hydroxylases (P4Hs), resulting in incomplete modification before extracellular release. Additionally, hydroxylation may hinder secretion capacity. To explore these hypotheses and verify the P4H activity of the enzymes, we decided to detect and compare hydroxylation levels in intracellular versus secreted collagen fractions by LC-MS/MS, expecting greater hydroxylation occupancy in intracellular pools due to prolonged enzyme interaction.

Mass spectrometry results demonstrated hydroxylation by both prolyl-4 hydroxylases and verified our hypothesis, by revealing significant elevated hydroxylation in intracellular fractions, particularly when MimiP4H was co-expressed (**Figure 14**). Notably, in SDA112 strain, more than 30% of the total intracellular peptides contained at least one (>1) hydroxylated proline, significantly more than the 2.8% exhibited by BaP4H (**SDA108 – Figure 14C**). Furthermore, SDA112 produced extensive proline hydroxylation events, with specific peptides exhibiting up to 10 hydroxyprolines simultaneously in the same peptide. This was the maximum possible number of hydroxyprolines based on the maximum available prolines of the digested peptides. This, in combination with the position of hydroxyprolines in the collagen sequence, suggests that MimiP4H is hydroxylating prolines irrespectively of their X or Y position on the Gly-X-Y motif (**Figure 14B**). Although high hydroxylation activity is desirable, uncontrolled and excessive hydroxylation can invoke negative effects in triple-helix stability. On the contrary, the maximum number of simultaneous hydroxyprolines detected on the BaP4H strain (SDA108) was five, while 96% of the intracellular or secreted hydroxylated peptides contained only 1 hydroxyproline, suggesting significantly low activity.

Interestingly, even though the secreted proteins showed reduced hydroxylation levels from 30% to 6.5%, SDA112-MimiP4H maintained similar distribution of the hydroxylated peptides, irrespectively of the fraction (**Figures 14E,F**). Unexpectedly, the intracellular production of collagen (SDA130) despite the presence of the highly efficient MimiP4H, produced hydroxylation levels closer to BaP4H (SDA108) than that of SDA112, reaching 4.4% of detected hydroxylated peptides (**Figure 14C**). SDA130 produced a lower coverage level (70%) of the sequence compared to the rest of the samples (≥ 86% coverage). This could mean that SDA130 had lower detection events, and the

hydroxylation levels were underestimated. Nevertheless, these results, in combination with the low production titres, constitute SDA130 an inefficient producer of collagen.

In conclusion, proteomics results revealed that extracellular fractions possess significantly less hydroxylation than the intracellular fractions. MimiP4H shows promising activity, but potentially exhibits uncontrolled hydroxylation, diverging from the native hydroxylation profile of collagen (**Figure 14B**). BaP4H exhibits significantly low activity despite high production levels (**Figure 11A**). SDA130 exhibited reduced MimiP4H activity compared to SDA112, failing to show advantages over the other strains.

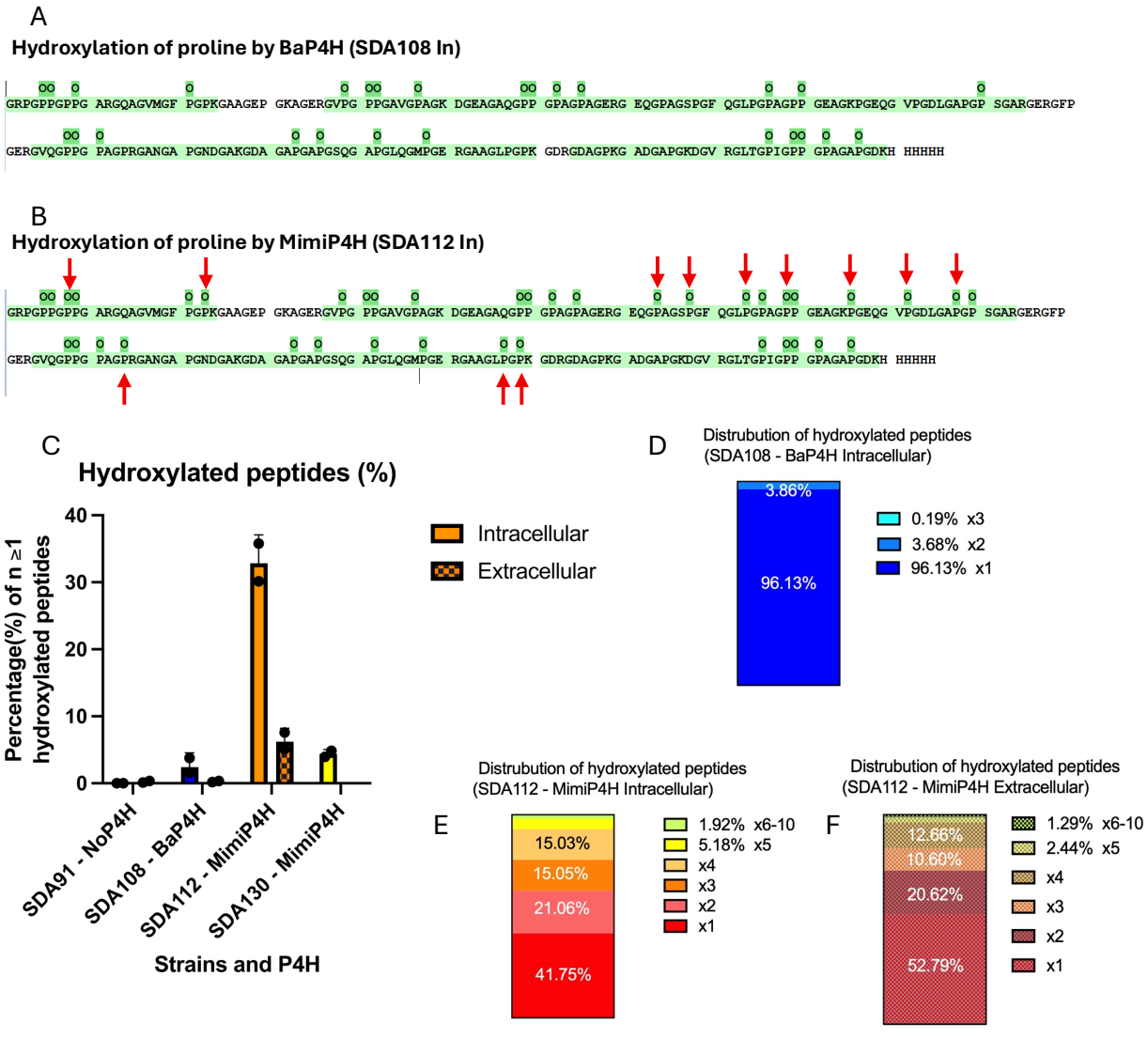


Figure 14 | Proteomics results of SDA91, SDA108, SDA112 and SDA130. *A, B shows the position of the detected hydroxyproline for BaP4H and MimiP4H respectively (green circle). The red arrow shows hydroxylation specific to MimiP4H compared to BaP4H. C, shows the percentage of peptides containing at least 1 hydroxyproline, intracellularly or extracellularly. D, E, F show the percentage of peptides containing 1x, 2x, 3x, 5, 6-10x hydroxyprolines simultaneously when detected.*

4.3 Successful triple-helix conformation and thermal stability for secreted rCollagen

To investigate the formation of triple-helical structures and the thermal stability of recombinant collagens, we performed circular dichroism (CD) spectrometry at 4°C. Triple-helical collagen presents characteristic CD spectra, with a negative peak near and below 200nm and a positive peak near 221nm. By testing different dilutions of bovine collagen type I, a universal concentration of 0.2mg/ml was selected as the most consistent for subsequent experiments (**Figure 15A**). Non-hydroxylated collagen CD (SDA91) produced a negative peak below 200nm but was lacking a distinct positive peak near 221nm. This phenotype corresponds to the formation of a random coil or partially assembled triple-helix. Experiments with higher concentrations of non-hydroxylated collagen, produced CD spectra with a positive peak at 221.5 nm, revealing the formation of a triple-helix (**Supplementary Figure 12**). Furthermore, purified secreted samples from either BaP4H or MimiP4H formed triple-helical structures, with no distinct differences between them or from the bovine standard (**Figure 15A-D**). Notably, the CD spectra of the intracellular fractions suggested the formation of α-helix instead of triple-helix (**Figure 16A-C)**. Intracellular fractions of the recombinant collagen still maintain the sequence of the secretion tag 5, which forms an alpha helix (**Supplementary Figure 13**). Therefore, a possible explanation could be that the secretion tag affects the detection of the triple-helical CD spectra or sterically hinders the assembly of the triple-helix. Another hypothesis includes the presence of co-purified α-helical proteins. Lampfish and starfish collagen type I were also investigated and showed only a negative peak at 200nm, suggesting the denaturation of collagen and random coil formation (**Supplementary Figure 12**).

To estimate the thermal stability of collagen samples, we conducted temperature-dependent scans at 221.5nm and measured the changes in the ellipticity. The calculated denaturation temperature (Td) of bovine collagen was at 39.9°C, which aligns to what is documented in literature. Surprisingly, the Td of all the secreted recombinant collagens, was estimated to be around 29.9°C with minor deviations.

In conclusion, these findings, combined with the mass spectrometry results, indicate that the hydroxylation of the secreted collagens is absent or limited for the BaP4H (SDA108) and MimiP4H (SDA112) strains, respectively. As a result, the thermal stability of the collagens remains similar to that of the non-hydroxylated collagen. Nevertheless,

the formation of triple-helical structures is possible and maintains a thermal stability of almost 30°C. Although this temperature is lower than native bovine collagen (40°C), it is comparable to or higher than that of the marine collagens (25-30°C). This suggests that the recombinant collagens may be suitable for biotechnological applications, particularly those where milder thermal stability is sufficient or even advantageous

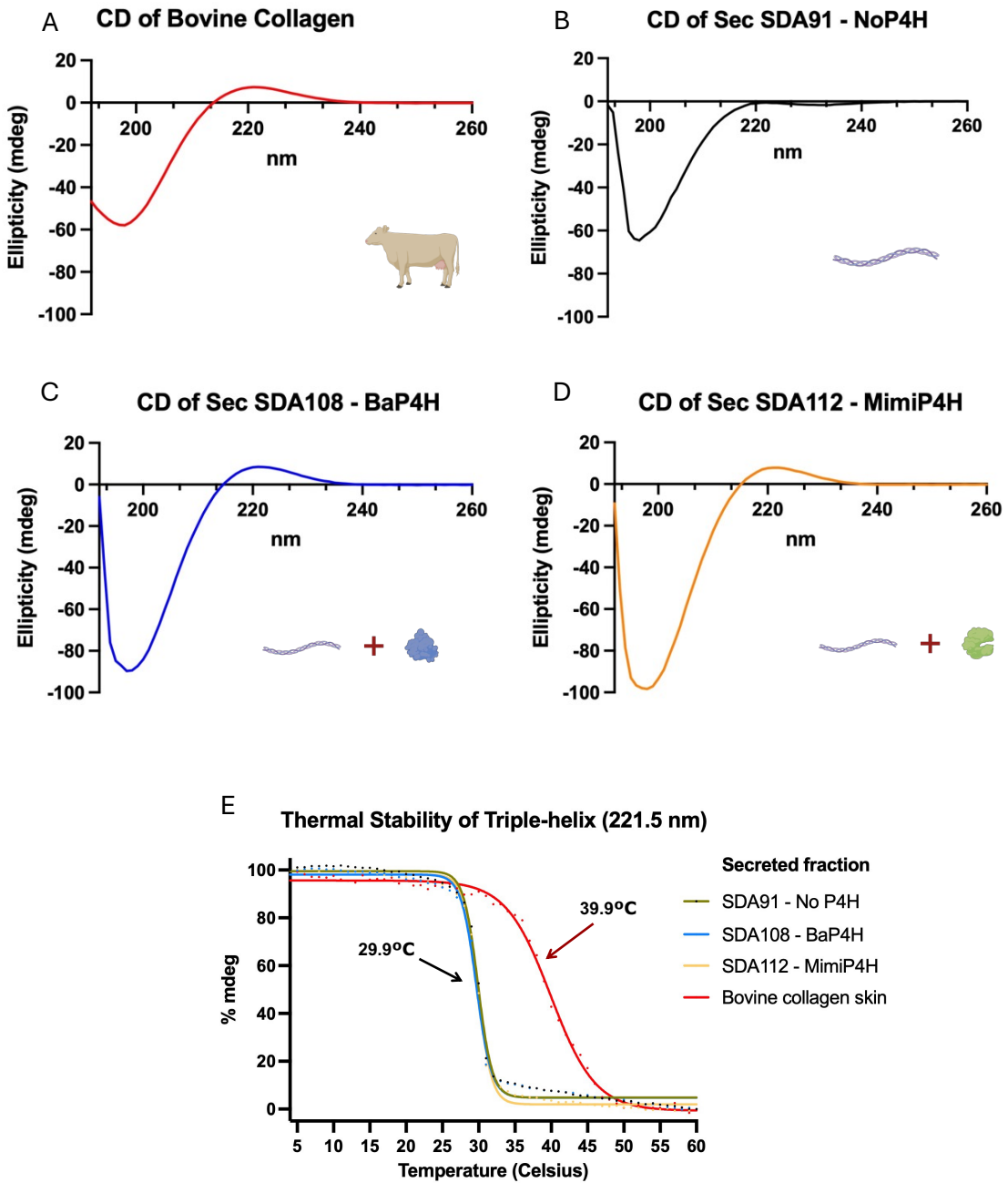


Figure 15 | Circular dichroism measurements prove the formation of triple-helix and the thermal stability of recombinant collagens. *A. Circular dichroism and ellipticity of bovine skin collagen. B-D Circular dichroism results for secreted fractions of SDA91, SDA108 and SDA112. E. Denaturation curve for all samples and bovine collagen. It reveals a melting point of 39.9 for the bovine collagen and 29.9 for recombinant collagen.*

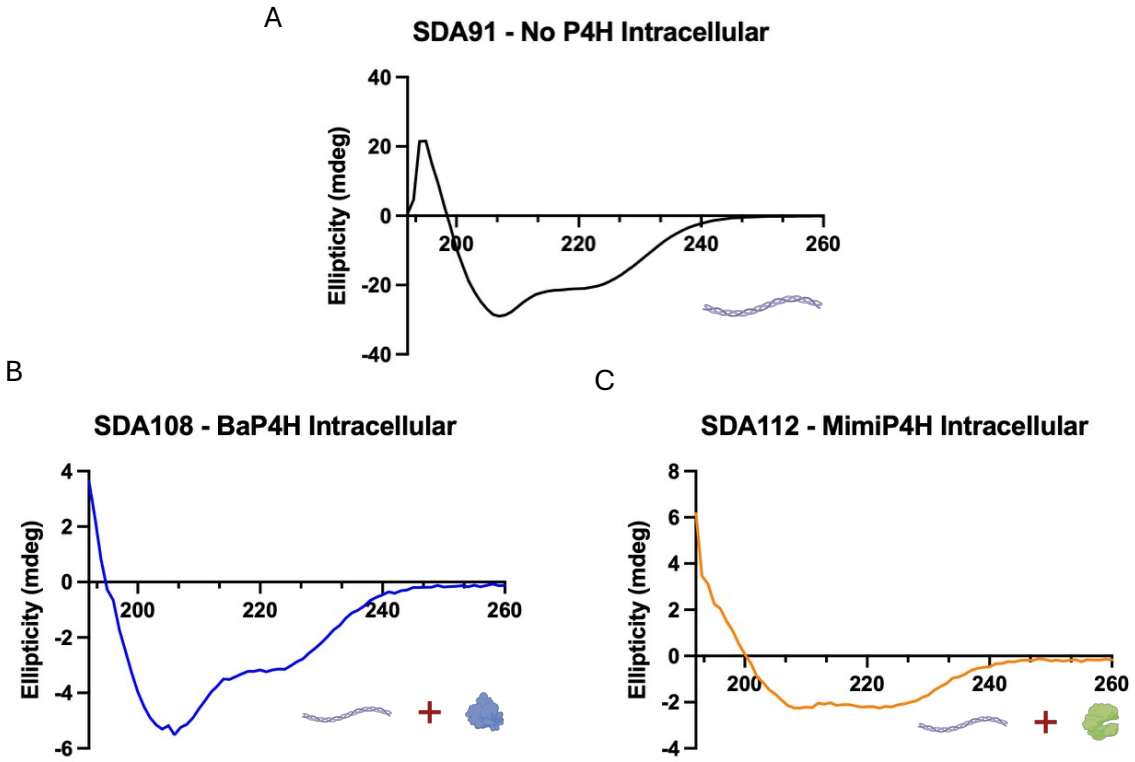


Figure 16 | Circular dichroism measurements of intracellular samples show the formation of α-helix. *All the samples show limited signal which corresponds to the formation of α-helix. This could be either due to the formation of α-helix by the secretion tag or the presence of contaminants*

Conclusion

In this study, we successfully demonstrated the first production and secretion of hydroxylated (bovine) collagen in *Bacillus subtilis.* By systematically screening 32 secretion peptides, we identified the best secretion peptides that enhance extracellular collagen secretion, particularly peptides SP5, SP19, SP26, and SP3. By progressively increasing the length of produced collagen, we highlighted the size-dependent challenges in collagen secretion and indicated that sequences containing the C-terminal propeptide may hinder production.

To introduce hydroxyproline in recombinant collagen, we co-expressed *Bacillus anthracis* P4H and *Mimivirus* P4H. Mass spectrometry revealed limited activity of *Bacillus anthracis P4H,* but relatively elevated activity in the intracellular fraction for *Mimivirus P4H*, with

>30% of collagen-derived peptides there showing relatively high levels of proline hydroxylation. Additionally, MimiP4H showed unconditional hydroxylation of prolines regardless of their position in the collagen sequence. Circular dichroism analysis revealed the formation of triple-helix for all secreted recombinant collagen despite their hydroxylation status, and their general stability up to 30°C. Interestingly, we showed that intracellular collagens did not form triple helices. Beyond collagen, we extended our secretion platform to other biopolymers, successfully producing the elastomers, resilin and elastin.

Overall, this work establishes a promising production system for hydroxylated collagen and opens the way for future food applications. Further research should focus on enhancing hydroxylation efficiency of P4Hs, scaling production and increasing collagen secretion titres.

Methods & Materials

1. Bacterial strains, Media, and Growth conditions.

For plasmid preparation, USER cloning and selection, *E. coli Dh5a* or *TOP10* stains (Thermo Fisher Scientific) were routinely grown in lysogeny broth (LB) media liquid cultures under shaking at 225-250 rpm or on LB agar plates at 37°C according to standard protocols. For protein production experiments, the extracellular protease-deficient *B. subtilis KO7S* strains were grown in either Terrific broth or in Cal18-2 media[61] (40 g/L yeast extract, 1.3 g/ $MgSO_4.7H_2O$, 50 g/L, maltodextrin (DE 13−17) Sigma-Aldrich, Saint Louis, MO, USA)), 20 g/L $NaH_2PO_4·2H_2O$, 6.7 mL/L Na_2MoO_4 stock solution (2.0 g/L), 6.7 mL/L trace metal solution (consisting of 4.48 g/L $MnSO_4·H_2O$, 3.33 g/L $FeCl_3·6H_2O$, 0.625 g/L $CuSO_4·5H_2O$, and 7.12 g/L $ZnSO_4·7H_2O$), and 100 µL/L Pluronic L-61 (Sigma-Aldrich, Saint Louis, MO, USA), then adjusted to pH = 6). When appropriate, the media were supplemented with antibiotics (100µg/mL ampicillin (Amp) or 50µg/mL kanamycin (Kan) for *E. coli* and 5µg/ml chloramphenicol, 5µg/ml kanamycin and 10µg/ml erythromycin for *B. subtilis KO7S*).

2. DNA manipulation, Plasmid and Strain Construction

All primers employed in this study were ordered from Integrated DNA Technologies (IDT) and are listed in Supplementary Table. The synthetic genes and fragments were purchased from either Integrated DNA Technologies or Twist Bioscience and always codon-optimized by the IDT codon-optimization tool. All plasmids and protein sequences infromation is found in the Supplementary tables.

For the construction of the *B. subtilis* collagen-secretion strains, the ProUSER2.0 integrative vectors were employed. The secretion peptides were native to *Bacillus subtilis* and were PCR amplified from the genome of the wild-type strain *Bacillus subtilis* 168. The codon-optimized gene of bovine collagen type I alpha I was synthesized according to the preferred codon usage of *B. subtilis* (IDT DNA) and cloned to integrative vectors with their respective secretion peptide partners. Specifically, 1µg of the ProUSER plasmid, pProUSER13C1B, that bears homology arms for the *B. subtilis glmS* genomic locus, was digested by mixing with 1µl of AsiSI (New England BioLabs, USA), 5µl CutSmart buffer and DNAse-free water to a final volume of 50µl. The reaction was performed at 37°C for 16 hours followed by inactivation at 80°C for 20 minutes. In the same reaction, 1µl of the Nt.BbvCI was added, and the mix was

incubated for 1 hour at 37°C followed by inactivation at 80°C for 20 minutes. The appropriate band that corresponds to the digested plasmid was purified by gel extraction (Macherey-Nagel) after gel electrophoresis of 1 hour at 100V. Then, 10- 20ng of the nicked plasmid were mixed with the appropriate PCR-amplified secretion tags and collagen protein (Phusion U) and subsequently USER cloning protocol was followed as described before.

For the episomal expression of the collagenous and P4Hs genes, plasmids pHT01 and pHT100 were selected. Codon-optimized genes for *B. anthracis* P4H and Mimivirus P4H, collagens and other polymeric genes, were synthesized and USER cloned after both vectors and genes were amplified with PCR (Phusion U). For most P4H constructs, a Hibit-tag was positioned the N-terminal for simultaneous detection of collagen and P4H expression. For the Hibit-tag addition, complementary oligos were synthesized. Larger and highly repetitive sequences were constructed with the USER assembly of gene fragments. The USER-ligated plasmids were verified by sequencing and transformed to *Bacillus subtilis KO7S glmS::ComKS* cells.

3. Transformation of Bacterial Cells

The transformation of E. coli DH5a or TOP10 Chemically competent strains (ThermoFischer Scietific) used for plasmid cloning was completedperformed according to standard protocols. Similarly, the transformation of parent strain Bacillus subtilis KO7S glmS::comKS andor genome-reduced was performed PG10 with the appropriate plasmid of USER-cloned PCR fragments after mannitol induction and overexpression of the ComKS genes of supercompetency was performed as described elsewhere[62] (and Chapter 4). The recovery period was adjusted to 2 hours.

4. Protein production in. *B. subtilis*

Preliminary small-scale expression experiments and comparison of signal peptides

Single colonies were picked and inoculated, depending on their number, in either 15ml preculture tubes containing 5mL LB, or in 24 deep-well plates containing 3mL LB, supplemented with the appropriate antibiotics. The cultures were grown O/N at 37°C and 225 rpm. The next day, the cultures were back-diluted to an OD of 0.1 in a new 24-deep well with 3-4 mL of either fresh TB or Cal18-2 media. Strains containing plasmids were supplemented with the appropriate antibiotics. Depending on the experimental purposes, the cultures were grown at different times and temperatures, while maintain a steady 225rpm shaking. For initial expression tests, as well as for the comparison of the secretion peptides,

a temperature of 20°C for 72 hours was selected whereas for the co-expression of collagens and P4Hs, 30°C for 24 hours was preferred.

Larger-scale protein production experiments

Protein purifications were performed in large flasks of 2L containing 0.5L of terrific broth, supplemented with 1mM of ascorbic acid, 0.01% anti-foam (Thermofischer) and 5-10 µg/mL of the appropriate antibiotic. The media was inoculated at OD_{600} 0.1, and the cultures were grown at 37°C and 225rpm for 2hours for constitutive expression or until OD_{600} reached 0.6-0.9 for induction with 1mM of IPTG. Then, the temperature was reduced to 30°C and the cultivation was stopped 24hours after inoculation. The supernatant was separated from cell debris by centrifugation at 7000g for 30 minutes at 4°C. Subsequently, the cell pellet was resuspended in PBS, transferred to 50ml falcon tubes, and precipitated again with centrifugation for the next steps of protein purification. Simultaneously, the supernatant was passed through a 0.22µm filter and then concentrated with an Amicon Stirred Cell and a 10kDA cut-off nitrocellulose membrane for each sample.

5. Protein purification

The cells were lysed by using 5ml of B-PER complete lysis buffer (ThermoFischer Scientific – Tris based) per 1g of cells, supplemented with protease inhibitor tablets. The lysate was separated from lysed cells by centrifuging at 14.000g, 4°C for 20 minutes and subsequently filtered through 0.22µm filters. The purification of his-tagged proteins from the lysate and production media was performed by affinity chromatography using HisTrap™ 5ml HP columns in ÄKTA Pure 25M (Cytiva). Each fraction (intracellular or secreted) had its own dedicated column to avoid contaminations with cytosol contaminants. Before purification, samples were supplemented with imidazole to reach a final concentration of 20mM. The buffer (A) for equilibration and washing of the affinity columns was composed of 40mM Tris-HCl, 300mM NaCl, 20mM imidazole, pH 7.6, and Buffer (B) for elution was similar but with 500mM of imidazole instead of 20mM. After the elution, fractions were analyzed by SDS-PAGE and western blotting and the ones containing the his-tagged proteins were pooled and dialyzed to 20mM Tris-HCl, 150mM NaCl, pH 7.4 with the Slide-A-Lyzer™ G3 Dialysis Cassettes, 10K MWCO (ThermoFischer Scientific) at 4°C. Afterwards, the purified proteins were aliquoted and stored at -80°C.

6. SDS-PAGE, Western blotting and signal intensity comparison

SDS-PAGE

Lysate and production media samples were mixed with the appropriate amounts of 4x Laemmli buffer (Bio-rad Laboratories), supplemented with 10x of reducing agent (ThermoFischer Scientific) and boiled at 95°C for 10 minutes. Subsequently, proteins were separated on a gradient (4–20%) mini-PROTEAN-TGX precast polyacrylamide gels (Bio-Rad Laboratories) and a PageRuler™ Plus Prestained Protein Ladder (ThermoFischer Scientific) was used as a molecular weight reference. Coomassie staining was performed with the use of InstantBlue Protein staining (ThermoFischer Scientific).

Western blotting

Proteins were transferred from electrophoresis gels to iBlot™ 2 regular or mini transfer nitrocellulose stacks by using the iBlot™ 2 gel transfer device. Depending on the molecular weight of the proteins of interest, the duration of blotting was different, ranging from 3min for low MW to 10min for higher MW proteins (21-25 volt). To visualize the total protein per lane, the membranes were incubated for 10-15 minutes with Ponceau S straining solution. After discarding the solution and washing with MQ water, the distinct red bands were visualized in a AI600 Imager (GE Healthcare) and saved for further analysis. Subsequently, the membranes were washed with 1x Tris-saline buffer solution- 0.1% Tween 20 (TBS-T) until no red bands were visible and incubated for 1hour with 5% (w/v) skim milk in TBS-T (Blocking), followed by the inoculation of the membranes with a 1:1000 dilution of the anti-His tag mouse primary antibody (05-949, Merck) in a fresh blocking solution for 16-18 hours at 4°C. After the overnight incubation and washing with TBS-T three times, the membranes were incubated with the anti-mouse IgG HRP secondary antibody (1:15000 dilution - GENA9310 Cytiva) for 1 hour and then washed again thrice with TBS-T. The chemiluminescent detection was performed with the Amersham™ ECL™ Prime kit (Cytiva) and the CCD AI600 Imager (GE Healthcate). All the incubations were on a rocking shaker.

Hibit blotting

Protein transfer from electrophoresis gels to the iBlot™ 2 membrane was performed using standard Western blotting procedures. After the transfer, the membranes were incubated in TBS-T buffer for 10 minutes on a rocker. A 200-fold dilution of the LargeBit protein in 1× Nano-Glo buffer was prepared and applied to the membranes. The membranes were then incubated for 1 hour at room temperature with gentle shaking. Following this, a 500-fold dilution of the Luciferase assay substrate was added directly to the LargeBit/buffer solution. The membranes were incubated in this mixture for 5 minutes at room

temperature.Finally, the membranes were imaged using the AI600 Imager with CCD detection.

Chemiluminescence signal intensity comparison

For the accurate relative comparison between bands, the analysis of the visualized membranes was performed in ImageLab software (Bio-rad Laboratories). For each membrate, the total protein content of each lane was calculated from the provided Poncaeu S staining images and a normalization factor was applied to the intensity levels obtained from the Western blots. Then, the mean intensity of all the bands in the same membrane was established and the ratio of sample normalized intensity to mean normalized intensity per membrane was calculated, enabling comparison. This was repeated for every biological replicate and the average ratio with standard error was used for the visuilation of the figures in Biorender.

7. ELISA

ELISA against the his-tagged proteins was performed after 50x dilution of culture supernatant with the assay's dilution buffer. The kit and protocol that it used was the "His Tag ELISA Detection Kit" from Genescript.

8. Proteomics

Proteomic sample preparation

The samples were processed using an SP3-workflow on a BioMek i7 liquid handling workstations (Beckman Coulter). 10µg of each sample were normalized to a volume of 150 µl. For samples with lower amount than 10µg 150 µL were taken without dilution. All samples were reduced in 10mM dithiothreitol at 56°C for 30min and alkylated in 20mM chloroacetamide at room temperature for 30min. 25 µL Hydrophobic and hydrophilic Sera-Mag™ SpeedBeads (50µg/µL, Carboxylate-Modified, Cytiva) were added to the samples. Proteins in the samples were precipitated on beads using acetonitrile (ACN, final concentration 70%), washed twice with 70% ethanol, once with 100% ACN and dried at room temperature. The beads were resuspended in digestion buffer (10mM CaCl2, 10% ACN in 50 mM HEPES) and proteins were digested with trypsin/LysC (0.5 µg, mass spec grade, Promega) for two hours at 37°C, followed by a second digestions step with trypsin (0.2 µg, mass spec grade, Promega) overnight at 37°C. The supernatants were transferred to new tubes, the beads were washed with 50 mM HEPES and the supernatants were combined.

Peptide concentrations were determined using Pierce™ Quantitative Fluorometric Peptide Assay (Thermo Scientific). The samples were purified using Pierce™ Peptide Desalting Spin Columns (Thermo Scientific), according to the manufacturer´s instructions, and evaporated to dryness in a SpeedVac concentrator.

LC-MS analysis

Peptides were dissolved in 3% ACN, 0.1% trifluoroacetic acid and 0.015% dodecyl-β-D-maltoside and injected on an Easy-nLC-1200 chromatography system coupled to an Orbitrap Fusion Lumos Tribrid Mass Spectrometer (both Thermo Scientific). The liquid chromatography system was equipped with an Acclaim Pepmap 100 C18 trap column (Thermo Scientific) and an in-house packed Reprosil-Pur C18 analytical column (75 µm × 350 mm, particle size 3 µm, Dr Maisch). The liquid chromatography flow was set to 300 nL/min, peptides were eluted with a gradient of 5% to 45% solvent B in solvent A over 48 min, then to 100% B over 2 min, and then kept at 100% B for 15 min. Solvent A was 0.2% formic acid in water, solvent B was 80% ACN and 0.2% formic acid in water.

The precursor ion mass spectra (MS1) were acquired at a resolution of 120 000 and an m/z range of 375-1500. Using a cycle time of 2 seconds the most abundant precursors with charges 2–7 were isolated with an m/z window of 1 and fragmented by higher-energy collisional dissociation (HCD) set to 30%. Fragment spectra (MS2) were recorded in the orbitrap at a resolution of 30 000 and dynamic exclusion was set to 10 sec.

Data Analysis

Data analysis was performed using Proteome Discoverer (ver 3.0, Thermo Fisher Scientific) The data was matched against bacillus subtilis swissprot database (4191 entries, April 2025) containing the submitted sequences of bovine collagen 20 kDa. Sequest HT was used as search engine with a precursor tolerance of 10 ppm and a fragment ion tolerance of 0.02 Da. Tryptic peptides were accepted with 2 missed cleavages. Methionine and proline oxidation were set as variable modifications and cysteine carbamidomethylation was set as fixed modification. IMP-ptmRS node was used for site localization of post translational modifications. Minora Feature Detector node was used for MS1 quantification.

9. Circular dichroism analysis

Native bovine collagen type I (C9791, Sigma-Aldrich), lumpfish and starfish (Mehdi Abdollahi's group, Chalmers university of Technology) were dissolved in 0.2M acetic acid and

diluted at a concentration of 1mg/ml, 0.2mg/mL and 0.1mg/ml and maintained overnight at 4°C for triple-helix formation. The absorption of the different concentrations was scanned from 190nm to 280nm with Chirascan CD spectrometer (Applied Photophysics) at 4°C, with 0.2M acetic acid as the baseline. The recombinant collagen samples and bovine collagen were diluted at 0.2mg/ml in TBS and stored at 4°C overnight. The absorption at 190-280nm was scanned by Chirascan at 4°C, with TBS as a baseline. The measurements were performed with a thermostated quartz cell of 1mm length, at a wavelength increment of 1nm and the results were derived from the mean of three consecutive scans. For the recombinant collagen, an additional purification batch was measured for reproducibility purposes. The native and recombinant collagens were subjected in thermal denaturation analysis by increasing the temperature from 4°C to 60°C at a rate of 1°C per min. Throughout the heating process, the circular dichroism (CD) signal intensity (mdeg) was measured at 221.5nm every minute in triplicates. The temperature where 50% of the triple-helix is denatured (Td) was estimated with the application of the Boltzmann sigmoidal equation and the extrapolation of V50.

Author Contributions

A.A., I.M, L.Y. conceived this study. A.A. designed the experiments with the guidance of I.M., C.J., S.I.J, M.A. and L.Y. Generation of data and analysis was performed by A.A. The manuscript was written by A.A. and reviewed by L.Y with contributions from other authors.

Acknowledgements

This work was funded by Novo Nordirsk Foundatio. We thank Johannes Fuchs, from Gothenburg University Facilities for the proteomics data generation and initial analysis.

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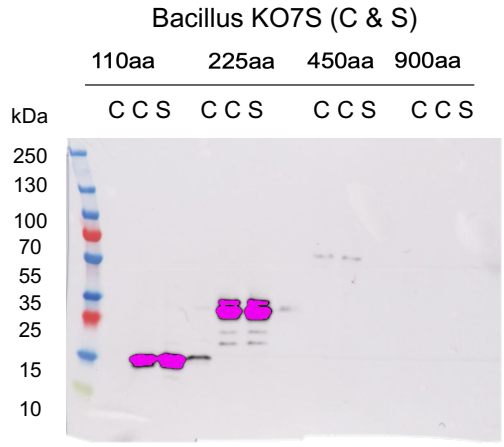
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Supplementary

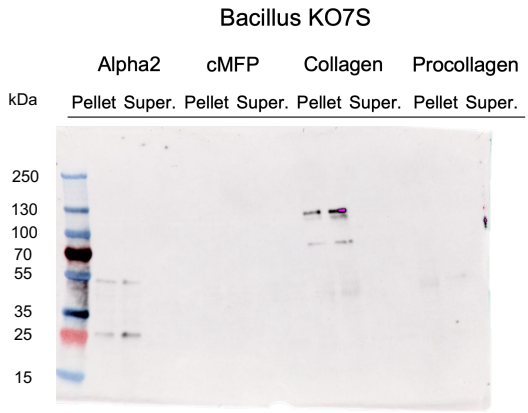
Amino Acid Frequencies of Tropocollagen type I alpha I			
Amino acid		**Count**	
Ala	A	122	11.6%
Arg	R	52	4.9%
Asn	N	11	1.0%
Asp	D	34	3.2%
Cys	C	0	0.0%
Gln	Q	30	2.9%
Glu	E	49	4.7%
Gly	G	346	32.9%
His	H	3	0.3%
Ile	I	9	0.9%
Leu	L	22	2.1%
Lys	K	38	3.6%
Met	M	7	0.7%
Phe	F	13	1.2%
Pro	P	241	22.9%
Ser	S	37	3.5%
Thr	T	17	1.6%
Trp	W	0	0.0%
Tyr	Y	4	0.4%
Val	V	17	1.6%

Amino acid frequencies of Collagen110			
Amino acid		**Count**	
Ala	A	14	13.0%
Arg	R	4	3.7%
Asn	N	0	0.0%
Asp	D	2	1.9%
Cys	C	0	0.0%
Gln	Q	4	3.7%
Glu	E	8	7.4%
Gly	G	34	31.5%
His	H	6	5.6%
Ile	I	0	0.0%
Leu	L	2	1.9%
Lys	K	4	3.7%
Met	M	0	0.0%
Phe	F	3	2.8%
Pro	P	22	20.4%
Ser	S	2	1.9%
Thr	T	0	0.0%
Trp	W	0	0.0%
Tyr	Y	0	0.0%
Val	V	3	2.8%

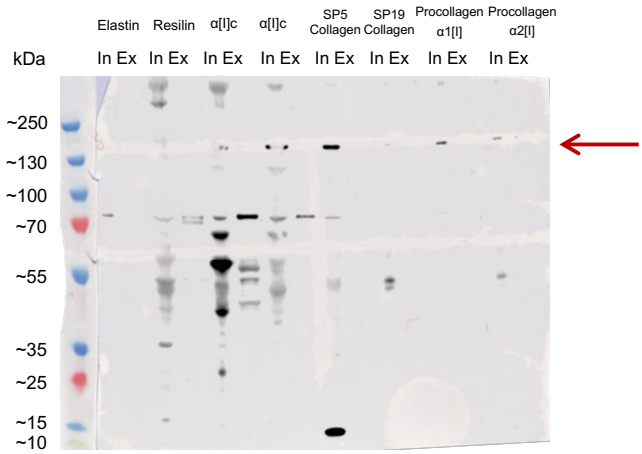
Supplementary Table 1 | *Amino acid frequencies of Native bovine collagen type I alpha I (left) and recombinant bovine collagen alpha I 10kDa (right)*



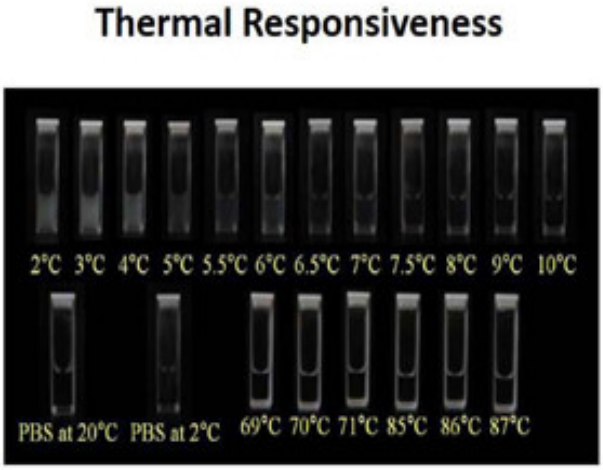
Supplementary Figure 1 | Contruction and producion of progressively extented collagens. *High exposure image of Western blot with progressively increasing collagen size. The shows the presence of 450aa collagen inside Bacillus subtilis cells, albeit in very low amounts compated to 110aa and 250aa. C: cell pellet, C': cell pellet replicate, S: Supernatant*



Supplementary Figure 2 | Western blotting of a2[I] with C-propeptide, cMFP, Collagen an Procollagen. *This western blot includes intracellular and extracellular samples of cultures producing the aforementioned proteins. It reveals that a2[I] with C-propeptide was in fact produced but in limited amounts and never secreted. Collagen also maintains the majority of its production intracellularly. cMFP was a synthetic construct of mussel food protein that never produced protein.*



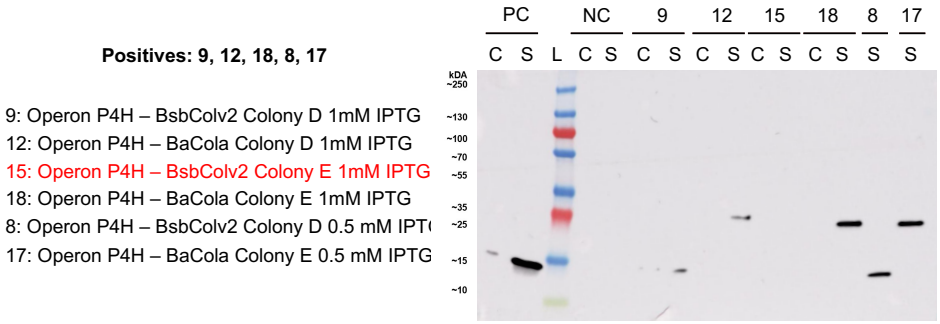
Supplementary Figure 3 | Western blotting of different strains. *This western blot includes intracellular and extracellular samples of cultures producing Elastin, Resilin, a1[I] with C-propeptide, cDNA bovine collagen (SP5, SP19) and codon-optimized procollagens. The red arrow indicated bands that potentially correspond to Procollagen bands. However, due to technical problems this western blot and subsequent westen blots produced significant backgrounds. Interestingly this experiment was performed in TB and exhibited differences to Cal-18 media, with some proteins secreting less quantities.*



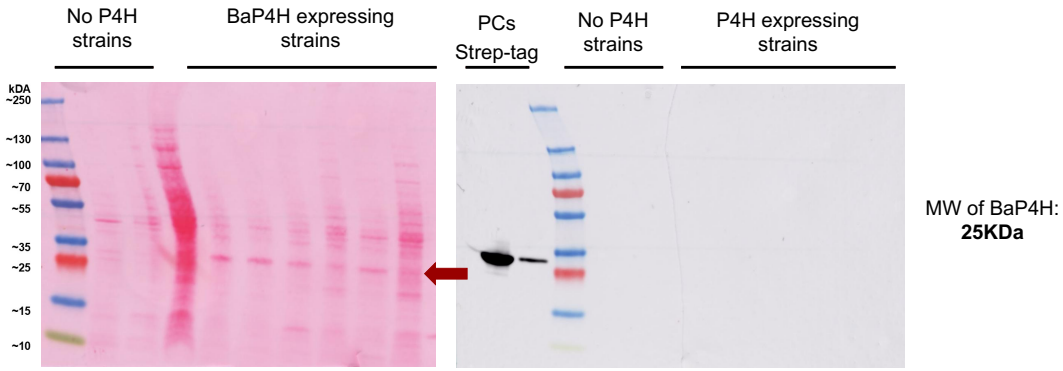
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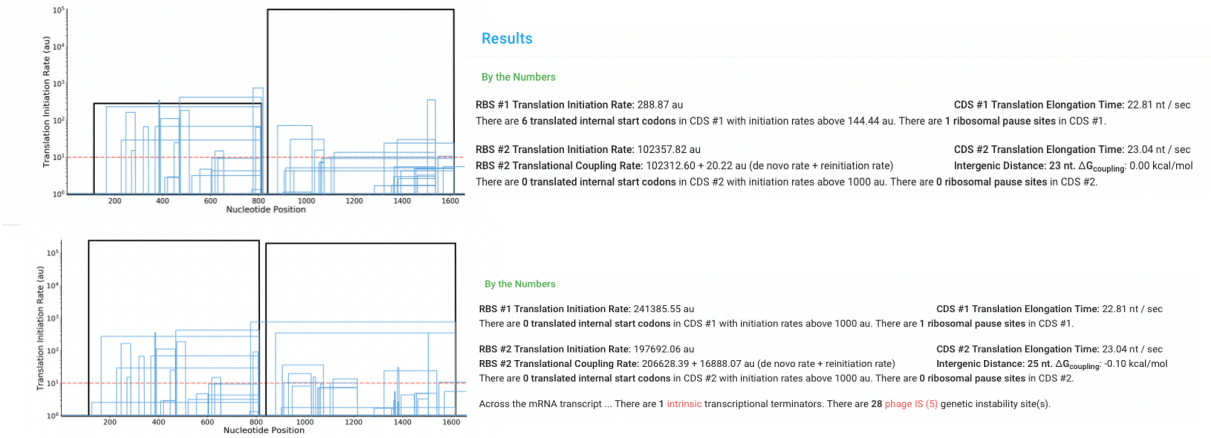
Supplementary Figure 4 | Depictions of resilin transition and opaqueness at low temperatures. Resilin's conformation is pH- and temperature-depedent. *It exhibits an upper critical solution temperature (UCTST) around 6 °C. Temperatures below UCST, promote conformational changes that lead to phase seperation and visible opaqueness as documented in the literature – Picture A. Picture B depicts recombinat resilin, produced, secreted and isolated with IMAC purification. During the O/N dialysis against TBS at 4 °C, the protein turned opaque while other proteins, such as non-hydroxylated collagen from BS091 strain, remained clear. This suggests that our recombinant variant might mimic the native resilin's structure, enabling possible applications.*



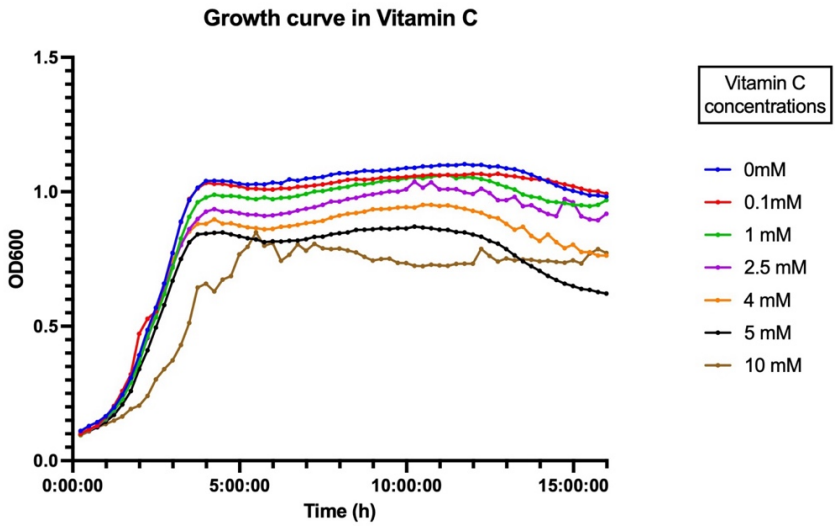
Supplementary Figure 5 | Western blotting of differernt pHT01 operon constructs expressing BaP4H and collagen fragments against His-tag (detection of collagen). *Operons expressing BaP4H and collagen 10kDa or 20kDa with secretion tag, where induced with 0.5mM or 1mM of IPTG. The results showed that induction with 0.5mM IPTG are more consistent. NC: Negative control – Supernatant of non-engineered strain. PC: Posive controls: BsbColv2 from previously. C: Cell pellet, S: Supernatant, L: Ladder.*



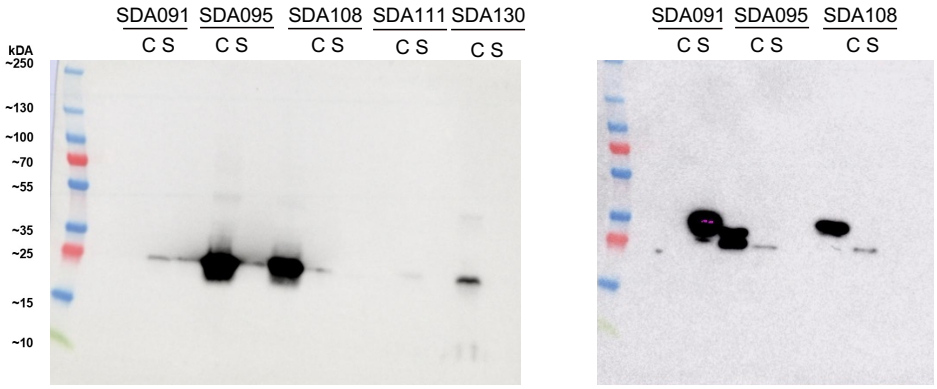
Supplementary Figure 6 | Western blotting of differernt constructs expressing BaP4H and collagen fragments (operon and genome) against Strep-tag II (detection of BaP4H). *Surprisingly anti Strep-tag antibody was not able to detect BaP4H (right). However, there are band in Ponceau stained membranes ar around 25kDa, that may indicate the production of BaP4H. Therefore, it is possible that detection of Strep-tag is stereochemically or otherwise, hindered.*



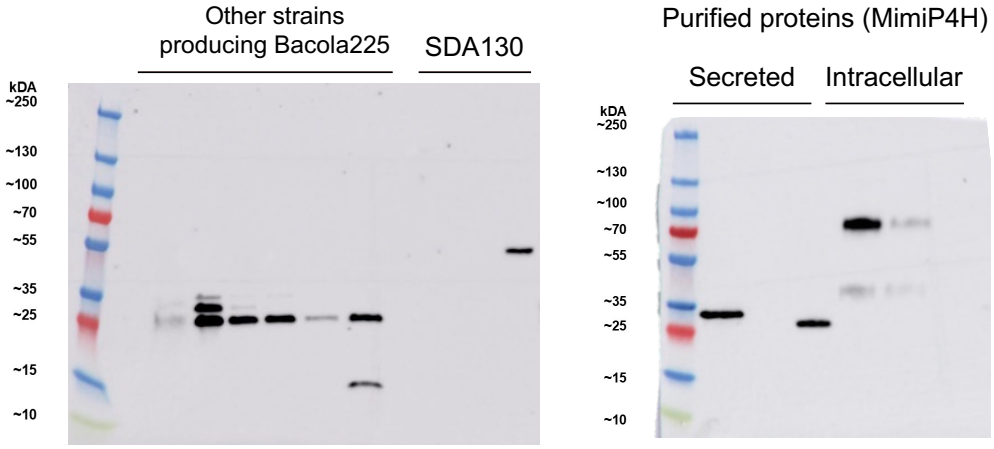
Supplementary Figure 7 | Prediction scores of the previous and novel RBSs for operon expression of P4Hs and collagen. *New designs increased considerably the estimated P4H translation rate from 289au to 241386au and the collagen translation rate from 102358 to 197692 .*



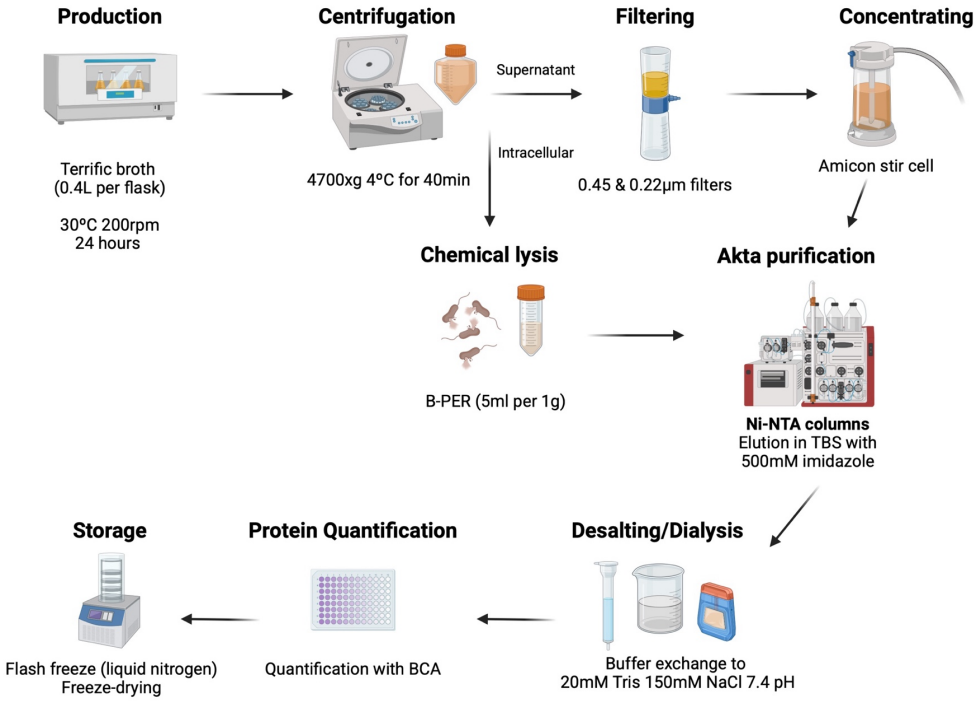
Supplementary Figure 8 | Dose-dependent growth curve of Bacillus subtilis KO7S in ascorbate. *Bacillus subtilis KO7S was grown in increased concentrations of ascorbate. Concentrations abobe 5mM affect growth from the exponential phase, while conentrations between 1mM to 5mM affect the final OD reached.*



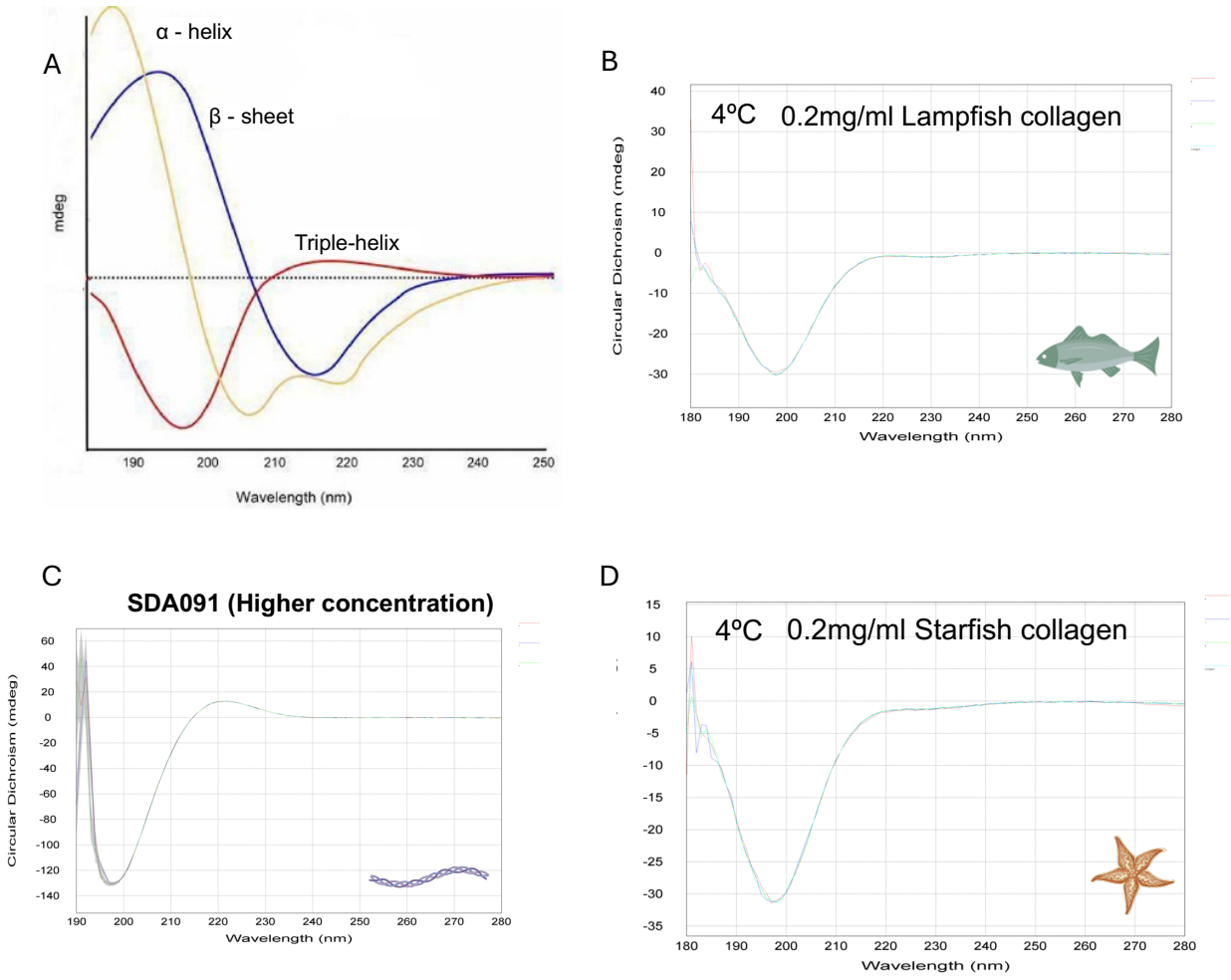
Supplementary Figure 9 | Verification of SDA095, SDA108, SDA111 and SDA130. *On the left is a Hibit blot that verified the expression of P4Hs. SDA095 and SDA108 produce intracellularly (C) BacP4H. SDA130 is expressing with RBS3* higher titres of MimiP4H than SDA111 with RBS1. On the right, there is a western blot that verified the production of collagen for the control (SDA091-pHT100 Bacola225) and SDA108, both intracellularly (C – Cell pellet) and extracellularly (S – Supernatant). In this western blot, SDA130, did not produce collagen.*



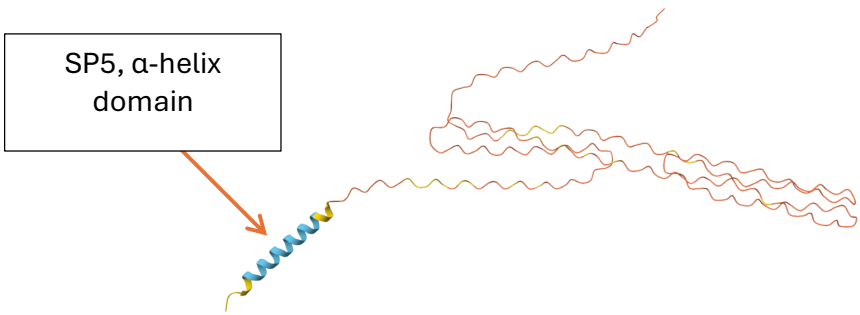
Supplementary Figure 10| Indications of hydroxylation or triple-helix formation. *In the first gel, western blotting of intracellular and extracellular proteins, indicated that collagen from SDA130, exhibited lower mobility compared to the same collagen from the other strains. In the second gel, intracellular collagen that was purified after the experiment of Figure 12, showed more prominent bands, in higher molecular weight than the secreted ones. This, along with the hypothesis that hydroxylation would be hyper inside the cell due higher exposure to P4H, indicate potential hydroxylation and/or triple-helix formation. Purification samples were left for some minutes at 4 degrees, which help the self-assembly of the triple-helix.*



Supplementary Figure 11| Graphical overview of the steps that were performed for recombiant collagen purification



Supplementary Figure 12 | Circular dichroism results. *A. Graphical representation of secondary structures signatures on Circular dichroism. B, D, Lampfish and starfish CD did not produce a positive peak at 221nm, but only low peak at <200nm. This indicates denaturation or degradation of collagen. C. In high concentration, non-hydroxylated recombinant collagen can form a triple-helical structure.*



Supplementary Figure 13 | Alpha-fold representation of SP5-Bacola225. The secretion signal SP5 has a α-helix domain that might affect circular dichroism results and/or triple-helix formation.

Table of plasmids

Entity Name	Alias	Usage
pBS001	proUSER13A1E	Backbone for USER cloning of inserts aimed to be transformed into bacillus
pBS002	proUSER13B4D	Backbone for USER cloning of inserts aimed to be transformed into bacillus
pBS003	proUSER13C1B	Backbone for USER cloning of inserts aimed to be transformed into bacillus
pBS004	proUSER13D2A	Backbone for USER cloning of inserts aimed to be transformed into bacillus
pBS005	proUSER13E3F	Backbone for USER cloning of inserts aimed to be transformed into bacillus
pBS006	proUSER13F4C	Backbone for USER cloning of inserts aimed to be transformed into bacillus
pBS007	proUSER13G2F	Backbone for USER cloning of inserts aimed to be transformed into bacillus
pBS008	pL001, SP1 - pProUSER13C1B-AmyESP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS009	pL002, SP2 - pProUSER13C1B-AprESP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS010	SP3 - pProUSER13C1B-BprSP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS011	SP4 - Not - pProUSER13C1B-EprSP-BsbCov2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS012	SP5 - pProUSER13C1B-MprSP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS013	SP6 - pProUSER13C1B-PelSP- BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS014	SP7 - pProUSER13C1B-PenPSP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS015	SP8 - pProUSER13C1B-PhoASP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS016	SP9 - pProUSER13C1B-PhoBSP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS017	SP10 - pProUSER13C1B-VprSP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS018	SP11 - pProUSER13C1B-XynASP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS019	SP12 - pProUSER13C1B-YbdNSP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS020	SP13 - pProUSER13C1B-YlqBSP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS021	SP14 - pProUSER13C1B-YncMSP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS022	SP15 - pProUSER13C1B-YoaWSP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS023	SP16 - pProUSER13C1B-YocHSP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS024	SP17 - pProUSER13C1B-YqxLSP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS025	SP18 - pProUSER13C1B-YvgOSP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS026	SP19 - pProUSER13C1B-YwaDSP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS027	SP20 - pProUSER13C1B-YweASP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)

pBS028	SP21 - pProUSER13C1B-YwtFSP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS029	SP22 - pProUSER13C1B-YxaLSP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS030	SP23 - pProUSER13C1B-NprBSP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS031	SP24 - pProUSER13C1B-SacBSP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS032	SP25 - pProUSER13C1B-SacCSP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS033	SP26 - pProUSER13C1B-AbnASP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS034	SP27 - pProUSER13C1B-BglSSP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS035	SP28 - pProUSER13C1B-YolASP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS036	SP29 - pProUSER13C1B-CsnSP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS037	SP30 - pProUSER13C1B-BsnSP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS038	SP31 - pProUSER13C1B-CwlOSP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS039	SP32 - pProUSER13C1B-EglSSP-BsbColv2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 110aa collagenous protein (BsbColv2)
pBS040	NoSP-BsbColv2-proUSER13C1B	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for intracelluar production of a 110aa collagenous protein (BsbColv2)
pBS041	SP3 - pProUSER13C1B-BprSP-BaCola225	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 225aa collagenous protein (Bacola225)
pBS042	SP5 - pProUSER13C1B-MprSP-BaCola225	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 225aa collagenous protein (Bacola225)
pBS043	SP19 - pProUSER13C1B-YwaDSP-BaCola225	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 225aa collagenous protein (Bacola225)
pBS044	SP26 - pProUSER13C1B-AbnASP-BaCola225	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of a 225aa collagenous protein (Bacola225)
pBS045	SP3 - pProUSER13C1B-BprSP-Procollagen	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of Procollagen
pBS046	SP5 - pProUSER13C1B-MprSP-Procollagen	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of Procollagen
pBS047	SP19 - pProUSER13C1B-YwaDSP-Procollagen	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of Procollagen
pBS048	SP26 - pProUSER13C1B-AbnASP-Procollagen	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of Procollagen
pBS049	SP3 - pProUSER13C1B-BprSP-Collagen	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of Collagen
pBS050	SP5 - pProUSER13C1B-MprSP-Collagen	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of Collagen
pBS051	SP19 - pProUSER13C1B-YwaDSP-Collagen	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of Collagen
pBS052	SP26 - pProUSER13C1B-AbnASP-Collagen	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of Collagen
pBS061	SP3 - pProUSER13C1B-BprSP-AlphaI With C	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of AlphaI with C telopeptide
pBS062	SP5 - pProUSER13C1B-MprSP-AlphaI with C	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of AlphaI with C telopeptide
pBS063	SP19 - pProUSER13C1B-YwaDSP-Alpha1 with C	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of AlphaI with C telopeptide

pBS064	SP26 - pProUSER13C1B-AbnASP-AlphaI with C	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of AlphaI with C telopeptide
pBS065	SP3 - pProUSER13C1B-BprSP-Alpha2 With C	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of Alpha2 with C telopeptide
pBS066	SP5 - pProUSER13C1B-MprSP-Alpha2 with C	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of Alpha2 with C telopeptide
pBS067	SP19 - pProUSER13C1B-YwaDSP-Alpha2 with C	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of Alpha2 with C telopeptide
pBS068	SP26 - pProUSER13C1B-AbnASP-Alpha2 with C	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of Alpha2 with C telopeptide
pBS069	SP3 - pProUSER13C1B-BprSP-Alpha1 without	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of Alpha1 withOUT C telopeptide
pBS070	SP5 - pProUSER13C1B-MprSP-AlphaI without	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of Alpha1 withOUT C telopeptide
pBS071	SP19 - pProUSER13C1B-YwaDSP-Alpha1 without	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of Alpha1 withOUT C telopeptide
pBS072	SP26 - pProUSER13C1B-AbnASP-AlphaI without	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of Alpha1 withOUT C telopeptide
pBS073	SP3 - pProUSER13C1B-BprSP-Alpha2 without	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of Alpha2 withOUT C telopeptide
pBS074	SP5 - pProUSER13C1B-MprSP-Alpha2 without	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of Alpha2 withOUT C telopeptide
pBS075	SP19 - pProUSER13C1B-YwaDSP-Alpha2 without	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of Alpha2 withOUT C telopeptide
pBS076	SP26 - pProUSER13C1B-AbnASP-Alpha2 without	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of Alpha2 withOUT C telopeptide
pBS077	pHT01_Pgrac100_empty	Episomical expression plasmid with a new promoter pgrac100
pBS078	SP5 - pProUSER13C1B-MprSP-BaCola450	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of Bacola450aa
pBS079	PHT01_Operon_P4H_BsbColv2	Episomical expression plasmid containing Strep-tag BacP4H and SP5-BsbColv2
pBS080	pHT01_Operon_P4H_Bacola225	Episomical expression plasmid containing Strep-tag BacP4H and SP5-Bacola225
pBS080	pHT01_Operon_P4H_Bacola225	Episomical expression plasmid containing Hibit-tag BacP4H and SP5-Bacola225
pBS081	pHT100_Operon_BacP4H_SP5_BsbColv2	Episomical expression plasmid containing promoter pgrac100, Strep-tag BacP4H and SP5-BsbColv2
pBS082	pHT100_Operon_BacP4H_SP5_Bacola225	Episomical expression plasmid containing promoter pgrac100, Strep-tag BacP4H and SP5-Bacola225
pBS082	pHT01_Pgrac100_Hibit_BacP4H_SP5_Bacola	Episomical expression plasmid containing promoter pgrac100, Hibit-tag BacP4H and SP5-Bacola225
pBS083	pTET_StrepBacP4H_NoLinker	Integration plasmid with pTet promoter (tet inducible).
pBS084	pProUSER13C1B-SP5-Bacola900	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of Bacola900aa
pBS085	pProUSER13C1B-SP5-WholeCollagenI	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of codon-optimised (pro)collagen I
pBS086	pProUSER13C1B-SP5-WholeAlpha2 His tag vol. 2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of codon-optimised (pro)collagen alpha 2 with a different his tag Nt Seq
pBS087	pProUSER13C1B-SP5-WholeAlpha2	Transformation of E.coli for USER assembly and tranfromation of B. subtilis for secretion of codon-optimised (pro)collagen alpha 2
pBS088	pHT100-SP5-Elastin	Episomical expression and secretion of human elastin through the pgrac100 promoter and Mpr(SP5) secretion peptide
pBS089	pHT100-SP5-Resilin	Episomical expression and secretion of D.melanogaster Resilin through the pgrac100 promoter and Mpr(SP5) secretion peptide

pBS091	pHT100-SP5-BsbColv2	Episomical expression and secretion of bovine BsbColv2 (110aa) through the pgrac100 promoter and Mpr(SP5) secretion peptide
pBS092	pHT100-SP5-Bacola225	Episomical expression and secretion of bovine Bacola225 (aa) through the pgrac100 promoter and Mpr(SP5) secretion peptide
pBS093	pHT100-SP5-Bacola450	Episomical expression and secretion of bovine Bacola450 (aa) through the pgrac100 promoter and Mpr(SP5) secretion peptide
pBS094	pHT100-Hibit-BacP4H	Episomical expression of BacP4H through the pgrac100 promoter and Hibit detection tag
pBS095	pHT100-Hibit-MimiP4H	Episomical expression of MimiP4H through the pgrac100 promoter and Hibit detection tag
pBS104	pHT100_Hibit_BacP4H_IntBacola225	Expression of the operon containing Hibit-BacP4H and intracellular Bacola225
pBS105	pHT100_Hibit_MimiP4H_SP5_Bacola225	Expression of the operon containing Hibit-MimiP4H and secreting Bacola225
pBS106	pHT100_Hibit_MimiP4H_IntBacola225	Expression of the operon containing Hibit-MimiP4H and intracellular Bacola225
pBS107	pHT100_NewRBS_Hibit_MimiP4H_SP5_Bacola225	Expression of the operon containing Hibit-MimiP4H and secreting Bacola225. Synthetic RBS1 and RBS2 were designed in RBS Calculator
pBS108	pHT100_NewRBS_Hibit_BacP4H_SP5_Bacola225	Expression of the operon containing Hibit-BacP4H and secreting Bacola225. Synthetic RBS1 and RBS2 were designed in RBS Calculator
pBS109	pHT100_NewRBS_Hibit_BacP4H_Int_Bacola225	Expression of the operon containing Hibit-BacP4H and Bacola225. Synthetic RBS1 and RBS2 were designed in RBS Calculator
pBS110	pHT100_NewRBS_Hibit_MimiP4H_Int_Bacola225	Expression of the operon containing Hibit-MimiP4H and Bacola225. Synthetic RBS1 and RBS2 were designed in RBS Calculator
pBS115	pHT-SP5-HisBacola	Expression and sectetion of Bacola225 with a N terminal His tag and TEV cleavage site under the pgrac100 promoter
pBS117	pHT-In-His-Bacola225	Expression and intracellular production of Bacola225 with a N terminal His tag and TEV cleavage site under the pgrac100 promoter
pBS118	pHT100_HibitBacP4H_TEV SP5 Bacola	Expression of operon containing Hibit-BacP4H and SP5-His-TEV-Bacola225
pBS119	pHT100_HibitMimiP4H_TEV SP5 Bacola	Expression of operon containing Hibit-m4H and SP5-His-TEV-Bacola225

Table of strains used in this study

Entity Name	Alias	Organism	Parent strain	Plasmids
SDA000	Bacillus subtilis KO7S	Bacillus subtilis		
SDA001	NoSignalPeptide BsbColv2	Bacillus subtilis	SDA000	pBS040
SDA002	SP1-BsbCovl2	Bacillus subtilis	SDA000	pBS008
SDA003	SP2-BsbCovl2	Bacillus subtilis	SDA000	pBS009
SDA004	SP3-BsbCovl2	Bacillus subtilis	SDA000	pBS010
SDA005	SP4-BsbCovl2	Bacillus subtilis	SDA000	pBS011
SDA006	SP5-BsbCovl2	Bacillus subtilis	SDA000	pBS012
SDA007	SP6-BsbCovl2	Bacillus subtilis	SDA000	pBS013
SDA008	SP7-BsbCovl2	Bacillus subtilis	SDA000	pBS014
SDA009	SP8-BsbCovl2	Bacillus subtilis	SDA000	pBS015
SDA010	SP9-BsbCovl2	Bacillus subtilis	SDA000	pBS016
SDA011	SP10-BsbCovl2	Bacillus subtilis	SDA000	pBS017
SDA012	SP11-BsbCovl2	Bacillus subtilis	SDA000	pBS018
SDA013	SP12-BsbCovl2	Bacillus subtilis	SDA000	pBS019
SDA014	SP13-BsbCovl2	Bacillus subtilis	SDA000	pBS020

SDA015	SP14-BsbCovl2	Bacillus subtilis	SDA000	pBS021
SDA016	SP15-BsbCovl2	Bacillus subtilis	SDA000	pBS022
SDA017	SP16-BsbCovl2	Bacillus subtilis	SDA000	pBS023
SDA018	SP17-BsbCovl2	Bacillus subtilis	SDA000	pBS024
SDA019	SP18-BsbCovl2	Bacillus subtilis	SDA000	pBS025
SDA020	SP19-BsbCovl2	Bacillus subtilis	SDA000	pBS026
SDA021	SP20-BsbCovl2	Bacillus subtilis	SDA000	pBS027
SDA022	SP21-BsbCovl2	Bacillus subtilis	SDA000	pBS028
SDA023	SP22-BsbCovl2	Bacillus subtilis	SDA000	pBS029
SDA024	SP23-BsbCovl2	Bacillus subtilis	SDA000	pBS030
SDA025	SP24-BsbCovl2	Bacillus subtilis	SDA000	pBS031
SDA026	SP25-BsbCovl2	Bacillus subtilis	SDA000	pBS032
SDA027	SP26-BsbCovl2	Bacillus subtilis	SDA000	pBS033
SDA028	SP27-BsbCovl2	Bacillus subtilis	SDA000	pBS034
SDA029	SP28-BsbCovl2	Bacillus subtilis	SDA000	pBS035
SDA030	SP29-BsbCovl2	Bacillus subtilis	SDA000	pBS036
SDA031	SP30-BsbCovl2	Bacillus subtilis	SDA000	pBS037
SDA032	SP31-BsbCovl2	Bacillus subtilis	SDA000	pBS038
SDA033	SP32-BsbCovl2	Bacillus subtilis	SDA000	pBS039
SDA034	SP3 BaCola225	Bacillus subtilis	SDA000	pBS041
SDA035	SP5 BaCola225	Bacillus subtilis	SDA000	pBS042
SDA036	SP19 BaCola225	Bacillus subtilis	SDA000	pBS043
SDA037	SP26 BaCola225	Bacillus subtilis	SDA000	pBS044
SDA042	SP3 With AlphaI	Bacillus subtilis	SDA000	pBS061
SDA043	SP5 With AlphaI	Bacillus subtilis	SDA000	pBS062
SDA044	SP19 With AlphaI	Bacillus subtilis	SDA000	pBS063
SDA045	SP26 With AlphaI	Bacillus subtilis	SDA000	pBS064
SDA046	SP3 With Alpha2	Bacillus subtilis	SDA000	pBS065
SDA047	SP5 With Alpha2	Bacillus subtilis	SDA000	pBS066
SDA048	SP19 With Alpha2	Bacillus subtilis	SDA000	pBS067
SDA049	SP26 With Alpha2	Bacillus subtilis	SDA000	pBS068
SDA050	SP3 WithOUT AlphaI	Bacillus subtilis	SDA000	pBS069
SDA051	SP5 WithOUT AlphaI	Bacillus subtilis	SDA000	pBS070
SDA052	SP19 WithOUT AlphaI	Bacillus subtilis	SDA000	pBS071
SDA053	SP26 WithOUT AlphaI	Bacillus subtilis	SDA000	pBS072
SDA054	SP3 WithOUT Alpha2	Bacillus subtilis	SDA000	pBS073
SDA055	SP5 WithOUT Alpha2	Bacillus subtilis	SDA000	pBS074
SDA056	SP19 WithOUT Alpha2	Bacillus subtilis	SDA000	pBS075
SDA057	SP26 WithOUT Alpha2	Bacillus subtilis	SDA000	pBS076
SDA058	SP3 Procollagen	Bacillus subtilis	SDA000	pBS045
SDA059	SP5 Procollagen	Bacillus subtilis	SDA000	pBS046
SDA060	SP19 Procollagen	Bacillus subtilis	SDA000	pBS047
SDA061	SP26 Procollagen	Bacillus subtilis	SDA000	pBS048

SDA062	SP3 Collagen	Bacillus subtilis	SDA000	pBS049
SDA063	SP5 Collagen	Bacillus subtilis	SDA000	pBS050
SDA064	SP19 Collagen	Bacillus subtilis	SDA000	pBS051
SDA065	SP26 Collagen	Bacillus subtilis	SDA000	pBS052
SDA070	Operon_pgrac_P4H_BsbColv2	Bacillus subtilis		pBS079
SDA071	Operon_pgrac_P4H_Bacola225	Bacillus subtilis		pBS080
SDA072	KO7S LinkP4H only	Bacillus subtilis	SDA000	pBS083
SDA073	KO7S NoLink_P4H only	Bacillus subtilis	SDA000	pBS083
SDA074	SP5-BsbColv-LinkP4H	Bacillus subtilis	SDA006	pBS083
SDA075	SP5-Bacola225-LinkP4H	Bacillus subtilis	SDA035	pBS083
SDA076	SP5-BsbColv-NoLinkP4H	Bacillus subtilis	SDA006	pBS083
SDA077	SP5-Bacola225-NoLinkP4H	Bacillus subtilis	SDA035	pBS083
SDA078	PG10-Operon P4H BsbColv2	Bacillus subtilis		pBS079
SDA079	PG10-Operon P4H Bacola225	Bacillus subtilis		pBS080
SDA080	PG10(2)-Operon P4H BsbColv2	Bacillus subtilis		pBS079
SDA081	PG10(2) -Operon P4H Bacola225	Bacillus subtilis		pBS080
SDA082	PG10-SP5-BsbColv2	Bacillus subtilis		pBS012
SDA083	SP5-Bacola450	Bacillus subtilis	SDA000	pBS078
SDA084	SP5-Bacola900	Bacillus subtilis	SDA000	pBS084
SDA085	PG10-SP5-Bacola450	Bacillus subtilis		pBS078
SDA086	PG10-SP5-Bacola900	Bacillus subtilis		pBS084
SDA087	pHT-SP5-Elastin	Bacillus subtilis		pBS088
SDA088	pHT-SP5-Resilin	Bacillus subtilis		pBS089
SDA090	pHT-SP5-BsbColv2	Bacillus subtilis		pBS091
SDA091	pHT-SP5-Bacola225	Bacillus subtilis		pBS092
SDA092	pHT-SP5-Bacola450	Bacillus subtilis		pBS093
SDA093	SP5-WholeCollagenA1	Bacillus subtilis	SDA000	pBS085
SDA094	SP5-WholeCollagenA2	Bacillus subtilis	SDA000	pBS087
SDA095	pHT-Hibit-BacP4H	Bacillus subtilis		pBS094
SDA096	pHT-Hibit-MimiP4H	Bacillus subtilis		pBS095
SDA108	pHT100_BacP4H_SP5_Bacola	Bacillus subtilis		pBS082
SDA109	pHT100_BacP4H_IntBacola225	Bacillus subtilis		pBS104
SDA110	pHT100_MimiP4H_SP5_Bacola	Bacillus subtilis		pBS105
SDA111	pHT100_MimiP4H_IntBacola225	Bacillus subtilis		pBS106
SDA112	pHT100_NewRBS_MimiP4H_SP5Bacola225	Bacillus subtilis		pBS107
SDA113	pHT100_NewRBS_BacP4H_SP5Bacola225	Bacillus subtilis		pBS108
SDA114	PG10_pHT100_BacP4H_SP5_Bacola	Bacillus subtilis		pBS082
SDA115	PG10_BacP4H_IntBacola225	Bacillus subtilis		pBS104
SDA116	PG10_MimiP4H_SP5_Bacola	Bacillus subtilis		pBS105

SDA117	PG10_MimiP4H_IntBacola225	Bacillus subtilis	pBS106
SDA118	PG10_NewRBS_MimiP4H_SP5Bacola225	Bacillus subtilis	pBS107
SDA119	PG10_NewRBS_BacP4H_SP5Bacola225	Bacillus subtilis	pBS108
SDA125	KO7S SP5-Hibit-Bacola	Bacillus subtilis	pSD553
SDA130	pHT100_NRBS_MimiP4H_InBacola225	Bacillus subtilis	pBS110
SDA133	KO7S_pHT-SP5-HisBacola	Bacillus subtilis	pBS115
SDA134	KO7S_pHT-In-His-Bacola225	Bacillus subtilis	pBS117
SDA135	KO7S_pHT-Hibit-BacP4H-SP5-Bacola225	Bacillus subtilis	pBS116
SDA136	KO7S_pHT-Hibit-BacP4H-In-Bacola225	Bacillus subtilis	pBS118
SDA137	KO7S_pHT-Hibit-MimiP4H-SP5-Bacola225	Bacillus subtilis	pBS119
SDA138	KO7S_pHT-Hibit-MimiP4H-In-Bacola225	Bacillus subtilis	pBS133

Table of strains used in this study

ID_Name	PRIMER SEQUENCE 5' – 3'
PRB1_BsbColv2_Fw	AGTGCUGGGTTTCCGGGACCAAAAGG
PRB2_BsbColv2_Rv	GGTGCGAUTTAATGGTGATGATGATGGTGCG
PRB7_Amye-tag1-U-fw	GGCGAUAGGAGGAATATACATGTTTGCAAAACGATTCAAA
PRB8_Amye-tag1-U-rv	AGCACUCGCAGCCGCCGGT
PRB9_Apre-tag2_U-fw	GGCGAUAGGAGGAATATACATGAGAAGCAAAAAATTGTGG
PRB10_Apre-tag2_U-rv	AGCACUCGCAGACATGTTGCTGAAC
PRB11_Bpr-tag3_U-fw	GGCGAUAGGAGGAATATACATGAGGAAAAAAACGAAAAAC
PRB12_Bpr-tag3_U-rv	AGCACUGGCTGCTCCCGGAAACAG
PRB13_Epr-tag4_U-fw	GGCGAUAGGAGGAATATACATGAAAAACATGTCTTGCAAA
PRB14_Epr-tag4_U-rv	AGCACUAGCGAGAGGGCCTATGGT
PRB15_Mpr-tag5_U-fw	GGCGAUAGGAGGAATATACATGAAATTAGTTCCAAGATTC
PRB16_Mpr-tag5_U-rv	AGCACUTGCCGGTACGCCAAAAGA
PRB17_Pel-tag6_U-fw	GGCGAUAGGAGGAATATACATGAAAAAAGTGATGTTAGCT
PRB18_Pel-tag6_U-rv	AGCACUCGCGCCAGCTGGAGTCAA
PRB19_Penp-tag7_U-fw	GGCGAUAGGAGGAATATACATGAAGTTGAAAACTAAAGCG
PRB20_Penp-tag7_U-rv	AGCACUGGCATGTGTTGAGTTGAAA
PRB21_Phoa-tag8_U-fw	GGCGAUAGGAGGAATATACATGAAAAAAATGAGTTTGTTT
PRB22_Phoa-tag8_U-rv	AGCACUGGCAAAGATTCCAGCTGTA
PRB23_Phob-tag9_U-fw	GGCGAUAGGAGGAATATACATGAAAAAATTCCCGAAGAAA
PRB24_Phob-tag9_U-rv	AGCACUGGCTTCAGGCACACTGCC
PRB25_Vpr-tag10_U-fw	GGCGAUAGGAGGAATATACATGAAAAAGGGGATCATTCGC
PRB26_Vpr-tag10_U-rv	AGCACUAGCTGCCTGAACGCCCGT
PRB27_Xyna-tag11_U-fw	GGCGAUAGGAGGAATATACATGTTTAAGTTTAAAAAGAAT
PRB28_Xyna-tag11_U-rv	AGCACUGGCGGTTGCCGAAAACAA
PRB29_Ybdn-tag12_U-fw	GGCGAUAGGAGGAATATACATGGTGAAAAAATGGCTTATT
PRB30_Ybdn-tag12_U-rv	AGCACUTGCCGAATACGTAAAGGTAGA
PRB31_Ylqb-tag13_U-fw	GGCGAUAGGAGGAATATACATGAAAAAAATCGGTTTATTG
PRB32_Ylqb-tag13_U-rv	AGCACUTGCTTGCTGTGCCGGGAA
PRB33_Yncm-tag14_U-fw	GGCGAUAGGAGGAATATACATGGCGAAACCACTATCAAAA

ID_Name	PRIMER SEQUENCE 5' – 3'
PRB34_Yncm-tag14_U-rv	AGCACUTGCCGCGGGTAAACCTGG
PRB35_Yoaw-tag15_U-fw	GGCGAUAGGAGGAATATACATGAAAAAGATGTTGATGTTA
PRB36_Yoaw-tag15_U-rv	AGCACUAGCTTCCCCTACATGGATAGT
PRB37_Yoch-tag16_U-fw	GGCGAUAGGAGGAATATACATGAAGAAGACGATTATGTCC
PRB38_Yoch-tag16_U-rv	AGCACUAGCTCCGAATGCAGTTGT
PRB39_Yqxi-tag17_U-fw	GGCGAUAGGAGGAATATACATGTTTAAGAAATTACTTTTA
PRB40_Yqxi-tag17_U-rv	AGCACUGGCATGTCCATCCAACGGA
PRB41_Yvgo-tag18_U-fw	GGCGAUAGGAGGAATATACATGAAACGTATTCGTATCCCA
PRB42_Yvgo-tag18_U-rv	AGCACUAGCAAAGGACAAAGGGGCA
PRB43_Ywad-tag19_U-fw	GGCGAUAGGAGGAATATACATGAAAAAGCTTTTGACTGTCAT
PRB44_Ywad-tag19_U-rv	AGCACUCGCGGCAGGGGTGACACT
PRB45_Ywea-tag20_U-fw	GGCGAUAGGAGGAATATACATGCTAAAAAGAACTTCATTC
PRB46_Ywea-tag20_U-rv	AGCACUAGCTTGGCCCGAAGGAAGT
PRB47_Ywtf-tag21_U-fw	GGCGAUAGGAGGAATATACATGGAAGAACGATCACAGCGC
PRB48_Ywtf-tag21_U-rv	AGCACUGGCGCCGACAGAGCCTGC
PRB49_Yxalk-tag22_U-fw	GGCGAUAGGAGGAATATACATGGTCAAGTCATTTCGTATGAAA
PRB50_Yxalk-tag22_U-rv	AGCACUAGCAGCTGTCGGCTGAAG
PRB51_Nprb-tag23_U-fw	GGCGAUAGGAGGAATATACATGCGCAACTTGACCAAGACA
PRB52_Nprb-tag23_U-rv	AGCACUGGCATGTGTTACAAAAACCA
PRB53_Sacb-tag24_U-fw	GGCGAUAGGAGGAATATACATGAACATCAAAAAGTTTGCA
PRB54_Sacb-tag24_U-rv	AGCACUCGCTTGAGTTGCGCCTCCT
PRB55_Sacc-tag25_U-fw	GGCGAUAGGAGGAATATACATGAAAAAGAGACTGATTCAA
PRB56_Sacc-tag25_U-r	AGCACUTGCCGAAAATGCCATAGT
PRB57_Abna-tag26_U-fw	GGCGAUAGGAGGAATATACATGAAAAAGAAAAAAACATGG
PRB58_Abna-tag26_U-rv	AGCACUTGCGGGAGCAGCAGAAGT
PRB59_Bgls-tag27_U-fw	GGCGAUAGGAGGAATATACATGCCTTATCTGAAACGAGTG
PRB60_Bgls-tag27_U-rv	AGCACUGGCAGTAGCAGTGACTGCA
PRB61_Yola-tag28_U-fw	GGCGAUAGGAGGAATATACATGAAGAAGAGAATTACATAT
PRB62_Yola-tag28_U-rv	AGCACUTGCTTTTGATGAATCAGTGAA
PRB63_Csn-tag29_U-fw	GGCGAUAGGAGGAATATACATGAAAATCAGTATGCAAAAA
PRB64_Csn-tag29_U-rv	AGCACUCGCCGTTTCGCTCATCATCAG
PRB65_Bsn-tag30_U-fw	GGCGAUAGGAGGAATATACATGACAAAAAAAGCATGGTTT
PRB66_Bsn-tag30_U-rv	AGCACUTGCTGAAGCTGCTGGCGCA
PRB67_Cwlo-tag31-U-fw	GGCGAUAGGAGGAATATACATGAGAAAGAGTTTAATTACA
PRB68_Cwlo-tag31-U-rv	AGCACUTGCAGTTTTACTTGTAAATGGGA
PRB69_Egls-tag32-U-fw	GGCGAUAGGAGGAATATACATGAAACGGTCAATCTCTATT
PRB70_Egls-tag32-U-rv	AGCACUTGCCGGCGAAGCTATCAT
PRB71_NoSP_BsbCov2_Fw	GGCGAUAGGAGGAATATACATGGGGTTTCCGGGACCAAAAG
PRB72_NoSP_BsbCov3_Fw	GGCGAUAGGAGGAATATACATGCCTGGAGAAAGAGGCGCTG
PRB73_553_pSEVA_seq_rv	CCGAGCGTTCTGAACAAATC
PRB74_2730_p3p-seq-fw2	TCTTGAAAGGAGGGATGCCT
PRB75_2074_glms_Check_Fw	TGAGCAGCCTGTTGTTATGC
PRB76_2075_glms_Check_Rv	TGGAAACGGGAAGAGATTGT
PRB77_BaCola_B_Fw	AGTGCUGGAAGACCTGGTCCACCTG

ID_Name	PRIMER SEQUENCE 5' – 3'
PRB78_BaCola_B_Rv	GGTGCGAUTTAATGGTGATGATGATGGTGTTTGTC
PRB79_CoBacola_Rv	CTGGTTCACCAGCGGCTC
PRB80_Seq_Bacola_Fw	GCAAGAGGAGAACGTGGTTTTCC
PRB81_AlphaI_Fw	AGTGCUGGACCGCCTGGTCTTGCAG
PRB82_WithAlphaI_Rv	GGTGCGAUTTAATGATGGTGGTGGTGGTGTAAG
PRB83_Scaf_W/O_AlphaI_Rv	GGTGGTGGTATCTACCTCCGTCGTGTGCC
PRB84_Tag_W/O_AlphaI_Rv	GGTGCGAUTTAATGATGGTGGTGGTGGTGGTATCTACCTCCG
PRB85_CoAlphaI_Rv	GTTCTCCCGGTGATCCCGGT
PRB86_Alpha2_Fw	AGTGCUGGCGCGAGAGGCCCAC
PRB87_WithAlpha2_Rv	GGTGCGAUTTAATGATGGTGGTGGTGGTGTTTG
PRB88_Scaf_W/O_Alpha2_Rv	GGTGGTGGGCCCGATAAAAGTCCCCGTC
PRB89_Tag_W/O_Alpha2_Rv	GGTGCGAUTTAATGATGGTGGTGGTGGTGGGCCCGA
PRB90_CoAlpha2_Rv	GAGCCCCTTGGTCACCGTG
PRB99_Wi_BovCola_Fw	AGTGCUGGCCAAGAGGAGGGCCAG
PRB100_Scaf_Wi_BovCola_Rv	GGTGGTGCAGGAAGCAGGCAGGGCC
PRB101_Tag_Wi_BovCola_Rv	GGTGCGAUTTAATGATGGTGGTGGTGGTGCAGGAAGCAG
PRB102_w/o_BovCola_Fw	AGTGCUGGCCAGTTGTCTTACGGCTATGATG
PRB103_w/o_Scaf_BovCola_Rv	GGTGGTGAGCCCGGTAGTAGCGGC
PRB104_w/o_Tag_BovCola_Rv	GGTGCGAUTTAATGATGGTGGTGGTGGTGAGCCCGGT
PRB105_Seq1_BovCola_Rv	TCAGCACCAGGAGCTCCCTC
PRB106_Seq2_BovCola_Rv	ACCAGGTTGGCCATCAGCAC
PRB107_Seq3_BovCola_Rv	GGGTCCCTGAGCTCCAGCTTC
PRB108_Seq4_BovCola_Rv	GGAAGCCAGGAGCACCAGC
PRB109_NativeWith_BovCola_Rv	GGTGCGAUTTACAGGAAGCAGGCAGGGC
PRB110_NativeWO_BovCola_Rv	GGTGCGAUTTAAGCCCGGTAGTAGCGGC
PRB111_Tag_Wi_BovCola_Fw	AGTGCUGGCCAAGAGGAG
PRB112_Tag_WO_BovCola_Fw	AGTGCUGGCCAGTTGTCTTAC
PRB113_HisTag_With_A1cDNA_Rv	GGTGCGAUTTAATGATGGTGGTGGTGGTGCAGGAAGCAGGCAGGGCC
PRB114_HisTag_WO_A1cDNA_Rv	GGTGCGAUTTAATGATGGTGGTGGTGGTGAGCCCGGTAGTAGCGGC
PRB115_HisTag_WO_A1_Rv	GGTGCGAUTTAATGATGGTGGTGGTGGTGGTATCTACCTCCGTCGTGTGCC
PRB116_HisTag_WO_A2_Rv	GGTGCGAUTTAATGATGGTGGTGGTGGTGGGCCCGATAAAAGTCCCCGTC
PRB117_Gibson_B225_Fw	CCTGGTCTTGCACACCATCATCATCACCATTAAATCGCAC
PRB118_Gibson_B225_Rv	TGAGGGGCCAGCTTCTCCTTTGTCTCCTGGTGCCCCAGC
PRB119_Gibson_Twist_Fw	GGAGACAAAGGAGAAGCTGGCCCCTCAG
PRB120_Gibson_Twist_Rv	GTGATGATGATGGTGTGCAAGACCAGGCGGTCCC
PRB121_Gibson_Back900_Fw	CACCATCATCATCACCATTAAATCGCAC
PRB122_Gibson_Back900_Rw	CTGATTCTCCAGGTGGTCCTGCAAGACCAGGCGGTCCC
PRB123_Gibson_BaCola900_Fw	CTGGTCTTGCAGGACCACCTGGAGAATCAGGTAG
PRB124_Gibson_Bacola900_Rv	GTGCGATTTAATGGTGATGATGATGGTGTAAGAAACAGG
PRB125_BacillusP4H_Fw	GAAAAGGGAGGTTCCGGTGGAGGGTCCGGCGGTACCAACAATAATCAAATCGGCGAAAAC
PRB126_BacillusP4H_Rv	GGTGCGAUTTATTTATACGTTCCGCGACGGAC

ID_Name	PRIMER SEQUENCE 5' – 3'
PRB127_StrepTag_BacillusP4H_Fw	GGCGAUAGGAGGAATATACATGTGGAGTCATCCTCAATTTGAAAAGGGGGGTTCCGGTGG
PRB128_NoLinkP4H_Fw	GGCGAUAGGAGGAATATACATGTGGAGTCATCCTCAATTTGAAAAGACCAAC
PRB129_rsbp_Check_Fw	CGGAGTTAGAAATTGCCTTGG
PRB130_rspb_Check_Rv	CATGGTTCAGGTCGGAAATG
PRB131_Ptet_Seq_Fw	GTATAATTAAAATAAGCTTGATCGTAGCG
PRB132_P4H_bicistronic_Fw	AAGGATCCUATGTGGAGTCATCCTCAATTTG
PRB133_P4H_Bicistronic_Rv	ACTCTAGCUATTATTTATACGTTCCGCGACG
PRB134_SP5_BsbColv_Bicistronic_Fw	AGCTAGAGUAGGAGGAATATACATGAAATTAGTTCCAAGATTC
PRB135_Bis_BsbColv2_Rv	ACGTCGUTAATGGTGATGATGATGGTGCG
PRB136_U_PHT01_Backbone_Fw	ACGACGUCCCCGGGG
PRB137_pHT01_P4H_Rv	AGGATCCTUCCTCCTTTAATTGGGAATTGTTATCC
PRB138_Split_PHT01_Fw	ACCCCGUAGAAAAGATCAAAGG
PRB139_Split_PHT01_Rv	ACGGGGUCTGACGCTC
PRB140_Bis_Bacola_Rv	ACGTCGUTTAATGGTGATGATGATGGTG
PRB141_SPN_Prop_Fw	AGTGCUGGCCAAGAAGAGGGTCAGGAG
PRB142_SPN_Prop_Rv	ACCTGGUAATCCGCGAGGG
PRB143_Collstart_Fw	ACCAGGUGAGCGGGGCCGCC
PRB144_Collstart_Rv	AGGTCTUCCATCTTGGCCCGCCGG
PRB145_ColBone_Fw	AAGACCUGGTCCACCTGGG
PRB146_Alpha2a_Fw	AGTGCUCAATCTCTTCAGGAAGCAACTGCG
PRB147_Alpha2a_Rv	AGCTCCUGCAACTCCAGGAAGACCAGCTGC
PRB148_Alpha2b_Fw	AGGAGCUCCAGGATTGCCG
PRB149_Alpha2b_Rv	ACGACTUCCAATCGGGCC
PRB150_Alpha2c_Fw	AAGTCGUGGCCCGTCTGGACCACCAGG
PRB151_Alpha2c_Rv	AGGGCCUGCTATGCCCA
PRB152_Alpha2l_Fw	AGGCCCUCCGGGCGCGAGAGGCCC
PRB153_Alpha2_FixHis_Rv	ATGATGAUGGTGTTTGAAACAGACCGGGC
PRB154_Alpha2_FixHis_Rv	ATCATCAUCACCATTAAATCGCACCTCAGCGT
PRB155_ELN1_Rv	AGCCGCUGCCGCAGCCGCGGC
PRB156_ELN2_Fw	AGCGGCUGCAGCAAAAG
PRB157_ELN2_Rv	AGGGGCUACCCCAACGC
PRB158_ELN3_Fw	AGCCCCUGGAATCGGTCCGGGAGG
PRB159_Histag_ProUSER_Rv	GGTGCGAUTTAATGGTGATGATGATGGTG
PRB160_Elastin3_pHT_Rv	ACGTCGUTTACTTTCTCTTTCTCCCACAGGCTT
PRB161_Elastin3_ProU_Rv	GGTGCGAUTTACTTTCTCTTTCTCCCACAGGCT
PRB162_ResilinN_Rv	AAGGCCGUCCCCCATTGCCA
PRB163_ResilinK-Fw	ACGGCCTUCTTCCTCATATGGCGCT
PRB164_ResilinK_Rv	AATCGGAUGGCTTCTGGTTCTGGC
PRB165_ResilinL_Fw	ATCCGATUCATATGGGGCTCCAGGGTC
PRB166_ResilinL_pHT_Rv	ACGTCGUTTAATAACGATAGCCGCTCCCATC
PRB167_ResilinL_ProU_Rv	GGTGCGAUTTAATAACGATAGCCGCTCCCATC
PRB198_Hibit_P4H_1	ATGGGATCAAGCGTGTCAGGATGGAGACTCTTCAAAAAGATAAGCTCT
PRB199_Hibit_P4H_2	ATCTTTTTGAAGAGTCTCCATCCTGACACGCTTGATCCCATAGGATCCTT
PRB200_User_BacP4H_Fw	AAGCTCUGGAATGACCAACAATAATCAAATCGGCG

ID_Name	PRIMER SEQUENCE 5' – 3'
PRB201_USER_BacP4H_Rv	ACGTCGUTTATTTATACGTTCCGCGACGGAC
PRB202_MimiP4H_Fw	AAGCTCUGGAATGAAGACCGTGACGATAATAACG
PRB203_MimiP4H_Rv	ACGTCGUTTAAGAGAACTTGCGCTCACG
PRB204_Hibit_SP5_Fw	AAGCTCUGGAATGAAATTAGTTCCAAGATTCAGAAAACAATG
PRB205_Hibit_InAmyE_Fw	AAGCTCUGGAATGGAAACGGCGAACAAATCG
PRB206_pHT_Hibit_Substitute	AGAGCTUATCTTTTTGAAGAGTCTCCATCCTGACACGCTTGATCCCATAGGATCCTTCCTCCTTTAATTGGGAATTGTTATCC
PRB209_His_TEV_1	GAAAATACAGATTTTCACCACTGCTATGGTGATGATGATGATGGCTGCTACCCATTAGAAGTCTCCTT
PRB210_His_TEV_2	TCTAATGGGTAGCAGCCATCATCATCATCACCATAGCAGTGGTGAAAATCTGTATTTTCAGGGTAGT
PRB215_pHT_BacP4H	AAGGATCCUATGTGGAGTCATCCTCAATTTG
PRB216_Bacola_pHT_Fw	AAGGATCCUATGGGAAGACCTGGTCCAC
PRB217_IntraBsbColv_Fw	AAGGATCCUGGGTTTCCGGGACCAAAAG
PRB218_Bacola750_Fw	ACACCAUCATCATCACCATTAAA
PRB219_Bacola750_Rv	ATGGTGUGGCTGCGTCGGGTAC
PRB226_pHT_SP5_Fw	AAGGATCCUATGAAATTAGTTCCAAGATTCAGAAAACAATGG
PRB240_Gibson_Hibit_Fw	TCGATTAGCTAAAGGAGGTATTCTAATGGGATCAAGCGTGTCAGG
PRB241_Gibson_RBS2_SP5_Rv	CATACCGAAGCCTCCTTTAGGGGATCTCTTATTTATACGTTCCGCGACGGACC
PRB242_Gibson_RBS2_SP5_Fw	GAGATCCCCTAAAGGAGGCTTCGGTATGAAATTAGTTCCAAGATTCAGAAAACAATG
PRB243_Spare_Gibson_Histag_Rv	TTAATGGTGATGATGATGGTGTTTGTCTC
PRB244_Spare_Gibson_His_tag_Fw	GGAGACAAACACCATCATCATCACCATTAAACGACGTCCCCGGG
PRB245_pHT_Hibit_Gibson_Rv	GATCCCATTAGAATACCTCCTTTAGCTAATCGAAATTGGGAATTGTTATCCGCTCAC
PRB246_GIbson_Bacola_Fw	GAGATCCCCTAAAGGAGGCTTCGGTATGGGAAGACCTGGTCCAC
PRB247_Gibson_MimiP4H_Rv	CATACCGAAGCCTCCTTTAGGGGATCTCTTAAGAGAACTTGCGCTCACG
PRB248_Int_Bacola_Fw	AGCTAGAGUAGGAGGAATATACATGGGAAGACCTGGTCCAC
PRB249_Operon_MimiP4H_Rv	ACTCTAGCUATTAAGAGAACTTGCGCTCACG
PRB256_Gibson_NewRBS2_Bacola	GAGATCCCCTAAAGGAGGCTTCGGTATGGGAAGACCTGGTCCA
PRB267_Gibson_CollagenIa_Fw	GGCCAACTTTCATATGGATACGACGAAAAAAGTACAG
PRB268_Gibson_CollagenaI_Rv	ACGTCGTTTAGGCGCGGTAGTATCTACCTCCGTCG
PRB269_Gibson_SP5_TEV	GTACTTTTTTCGTCGTATCCATATGAAAGTTGGCCCTGGAAATATAAATTTTCATGGTG
PRB275_NewRBS_pHT_Rv	AGAGCTUATCTTTTTGAAGAGTCTCCATC
PRB276_NewRBS2_Fw	AGATCCCCUAAAGGAGGCTTC
PRB277_BacP4H_User_NewRBS_Rv	AGGGGATCUCTTATTTATACGTTCCGCGACGGACCCAC